CEBU INSTITUTE OF TECHNOLOGY
UNIVERSITY

# AIRWise: Smart Room Responsive Remote-Control System for Thermal Comfort


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# INTRODUCTION

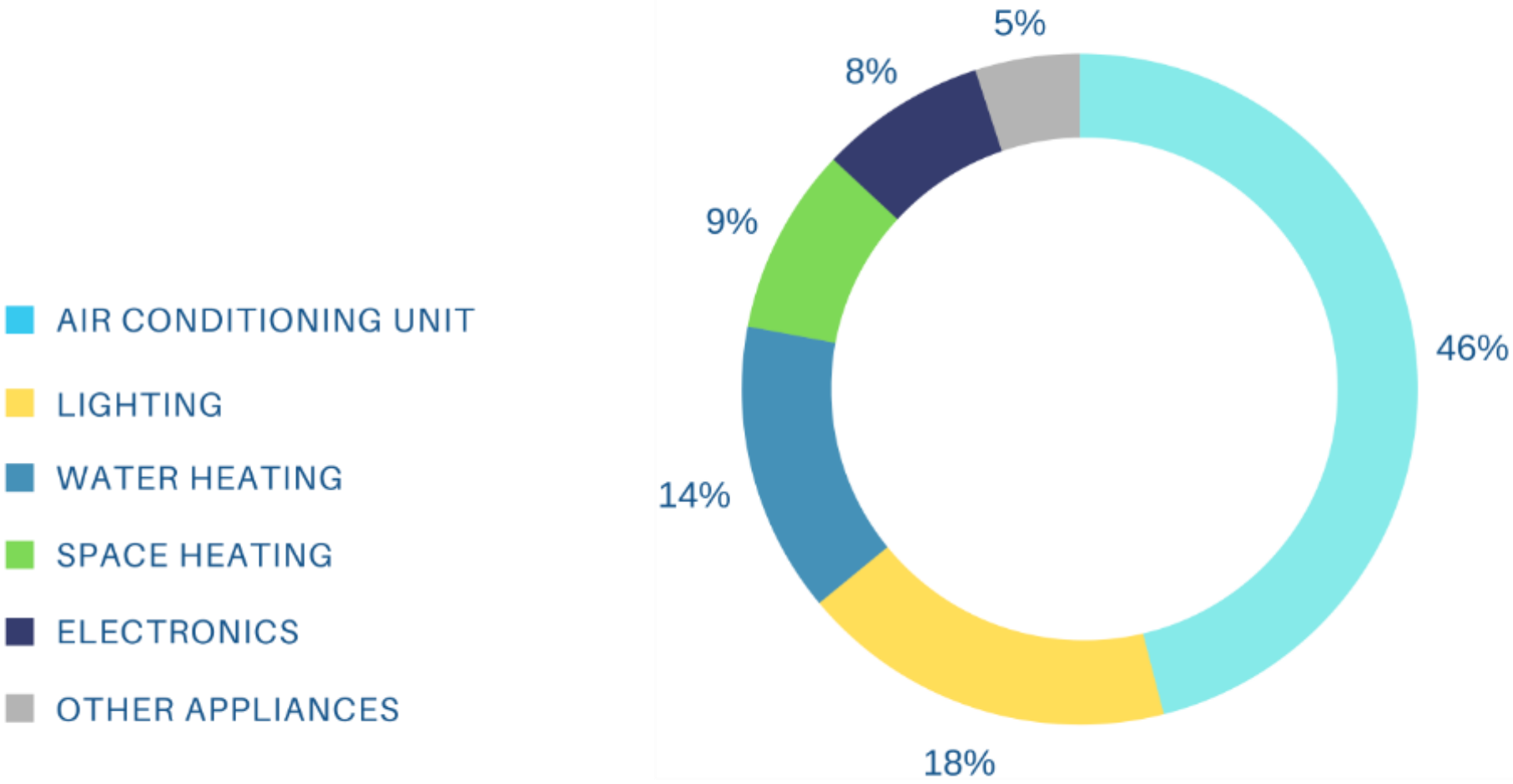
## Improper and Inefficient Usage of an Aircon

- Setting temperature far below outdoor conditions
- Neglecting swing, fan, and other mode settings
- Constantly switching the aircon **ON** and **OFF** rather than using proper settings
- Leaving the aircon **ON** when no one is using




[1] M. Hong, K. Wang, and T. Chen, "Smart Air-Conditioning Control Using IoT and Occupancy Detection," *IEEE Internet of Things Journal*, vol. 8, no. 4, pp. 2953–2965, 2021.
[2] International Energy Agency, "The Future of Cooling: Opportunities for energyefficient air conditioning," IEA, 2018. [Online]. Available: https://www.iea.org/reports/the-future-of-cooling
[3] A. Alhussein, G. Muhammad, M. S. Hossain, and S. U. Amin, "Cognitive IoT-Cloud Integration for Smart Remote Monitoring of AC Systems," *IEEE Access*, vol. 7, pp. 21090–21100, 2019.

# Electricity Consumption Breakdown




[1] M. Hong, K. Wang, and T. Chen, "Smart Air-Conditioning Control Using IoT and Occupancy Detection," *IEEE Internet of Things Journal*, vol. 8, no. 4, pp. 2953–2965, 2021.
[2] International Energy Agency, "The Future of Cooling: Opportunities for energyefficient air conditioning," IEA, 2018. [Online]. Available: https://www.iea.org/reports/the-future-of-cooling
[3] A. Alhussein, G. Muhammad, M. S. Hossain, and S. U. Amin, "Cognitive IoT-Cloud Integration for Smart Remote Monitoring of AC Systems," *IEEE Access*, vol. 7, pp. 21090–21100, 2019.

# PROBLEM

Air-conditioning units operate manually without adapting to room conditions, leading to energy waste and improper thermal comfort

# GENERAL OBJECTIVE

Design a smart, room-responsive remote-control system for air conditioning unit that automatically adjusts mode settings, fan speed, swing, and temperature based on the room's ambient conditions

# SPECIFIC OBJECTIVES

- Create a network that collect and monitor room ambient conditions using a set of wireless sensor nodes for data transmission
- Develop an intelligent decision system capable of determining and adjusting optimal air conditioning settings based on data collected from the wireless sensor nodes
- Design a wireless control hub that transmit the computed air-conditioning control settings from the intelligent decision system to the aircon

# PURPOSE OF THE STUDY









# PURPOSE OF THE STUDY

Building managers and maintenance, Users, Technology developers, Future researchers

# DEFINITION OF TERMS

**AIRWise:** an automated intelligent remote control for a wise indoor environment.

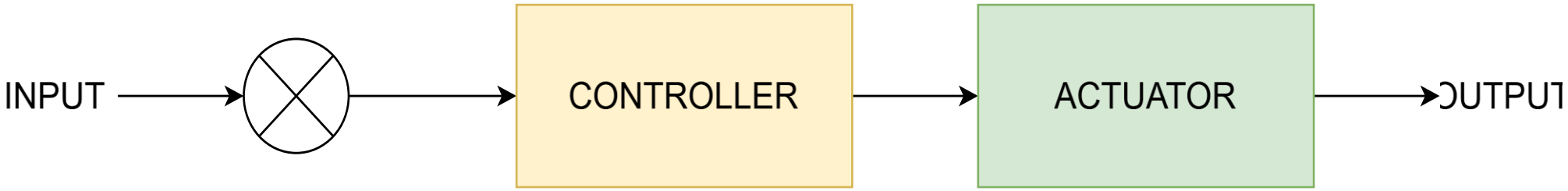
**AC (Aircon)**: a system for cooling indoor places.

**Sensor node:** a device that detects motion, humidity, and temperature in the system.

# THEORIES OUTLINE

1. CONTROL SYSTEM
2. WIRELESS COMMUNICATION
3. MODE AND SETTINGS OF AN AIRCON
4. WORKING PRINCIPLES OF SENSORS
5. THERMAL COMFORT

# THEORY
## CONTROL SYSTEMS




Responsible for regulating the operation of a device based on sensor data and feedback

https://electronicscoach.com/open-loop-control-system.html

# THEORY
## WIRELESS CONTROL SYSTEMS

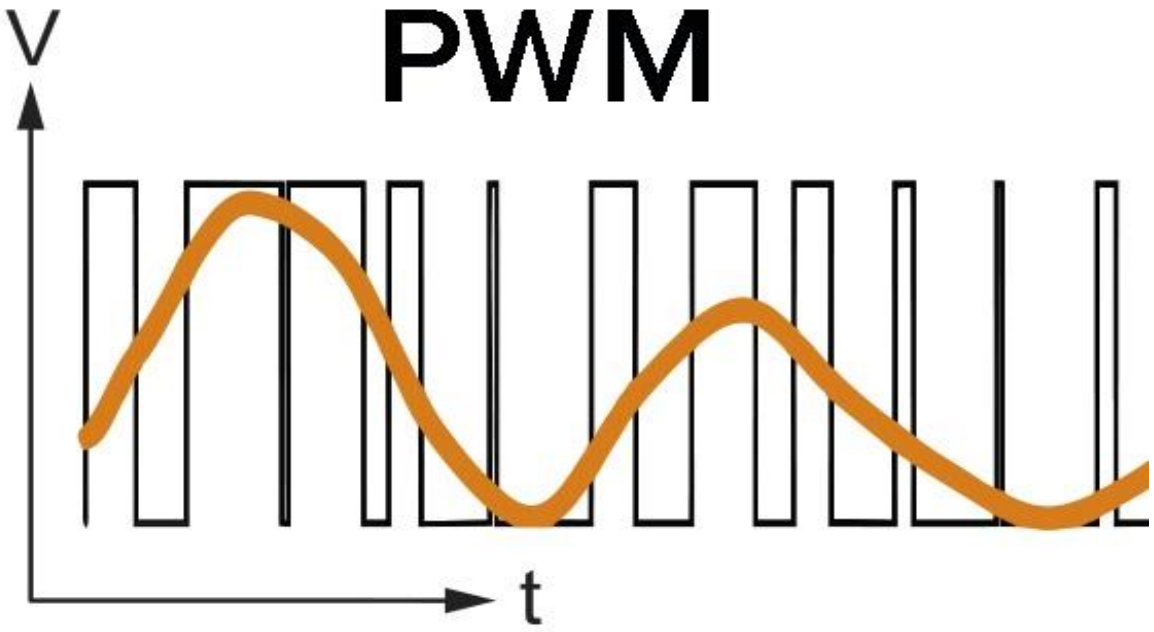



Wireless Communication




https://electronicscoach.com/open-loop-control-system.html

# THEORY
## INFRARED COMMUNICATION

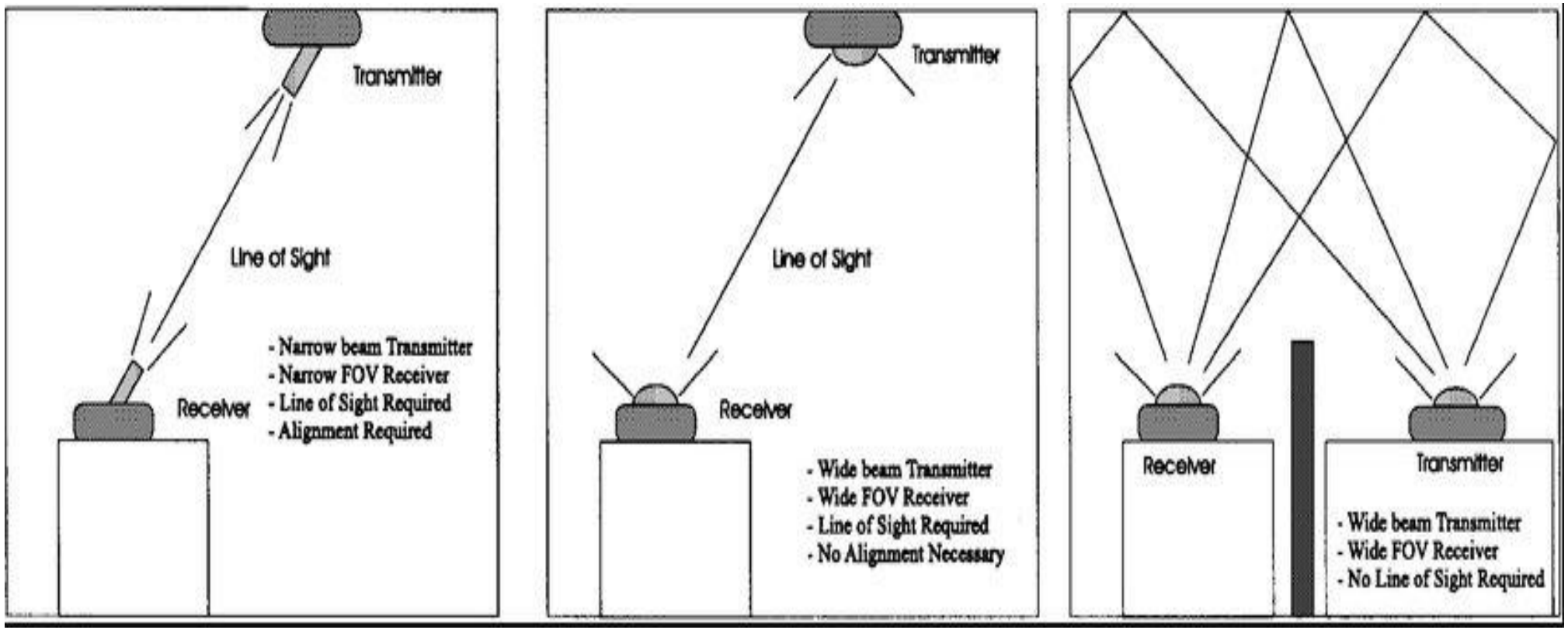





IR transmitted through a process of electromagnetic radiation

https://learn.sparkfun.com/tutorials/ir-communication/all

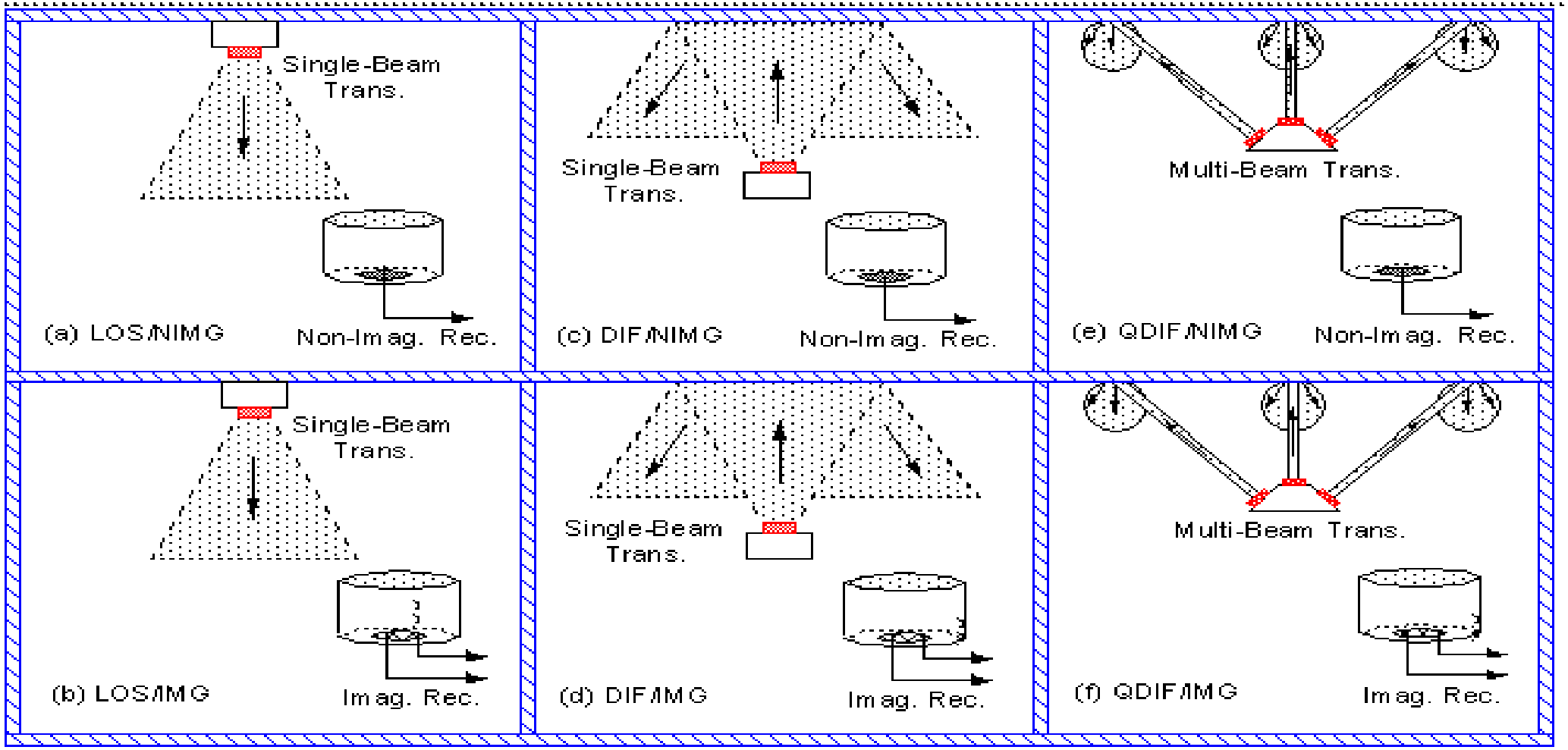
# THEORY
## INDOOR IR TRANSMISSION




https://www.researchgate.net/profile/Mr-Pakravan/publication/220232133/figure/fig1/AS:669947986452482@1536739455164/Illustration-of-some-of-common-indoor-infrared-transmission-conR-gurations-The-directed.pbm

# THEORY

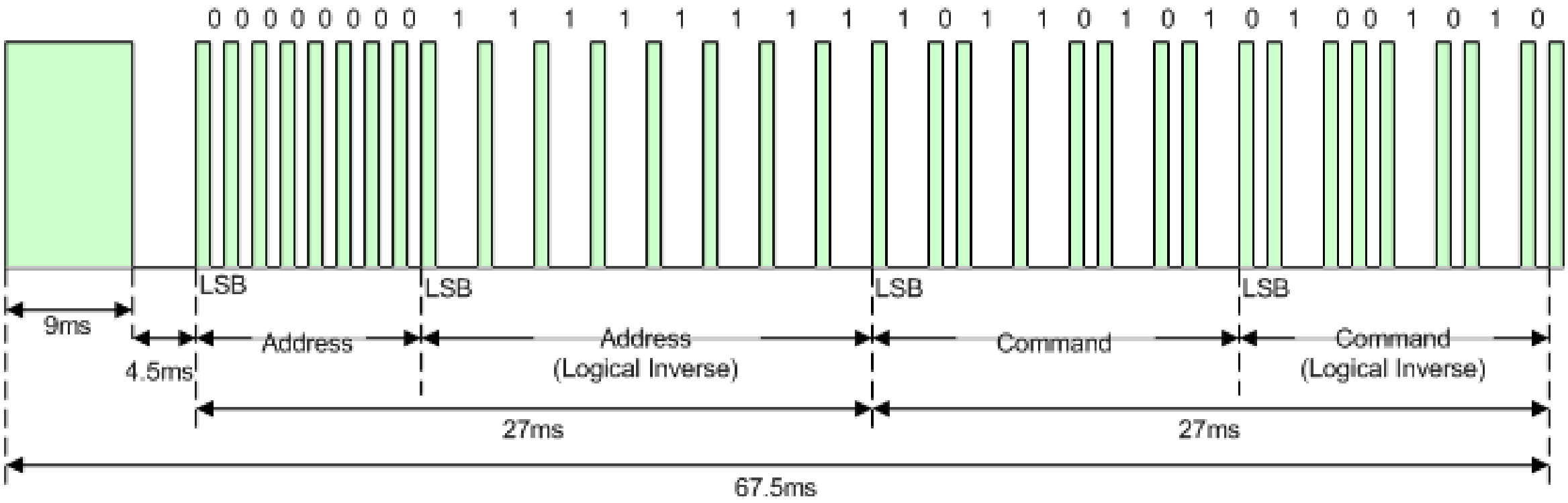
## CLASSIFICATION OF IR TRANSMISSION




http://www.wirelesscommunication.nl/reference/chaptr01/wrlsnfrr/inftrans.htm

# THEORY

## NEC PROTOCOL




https://exploreembedded.com/wiki/NEC_IR_Remote_Control_Interface_with_8051

# THEORY

## NEC PROTOCOL

Time representation of NEC encoding

| Start | Space | Address Checksum | Data Checksum | Stop |
|---|---|---|---|---|
| 9 ms | 4.5 ms | 27 ms | 27 ms | 562.5 us |

*https://medium.com/blueeast/nec-transmission-protocol-for-air-conditioners-67c8a698e633*

# THEORY

## NEC PROTOCOL

Hex/Time representation of NEC encoding

| | | START | SPACE | ADDRESS CHECKSUM | DATA CHECKSUM | | | | | | Checksum |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Temp | Swing | Fan | Mode | Timer | On/Off | |
| Example | Temp = 24<br>Swing = ON<br>Fan = ON<br>Mode = Cool<br>Timer = ON<br><br>Power = ON | 9 ms | 4.5 ms | 11 DA 27 | 30 | E7 | A0 | 0x30 | C3 | 0x40 | 562.5 us |
| | | | | 27 ms | 27 ms | | | | | | |

*https://medium.com/blueeast/nec-transmission-protocol-for-air-conditioners-67c8a698e633*

# THEORY

## MODE SETTINGS OF AN AIRCON














- **Cool Mode**
- **Fan Mode**
- **Dry Mode**
- **Auto Mode**
- **Heat Mode**

http://www.wirelesscommunication.nl/reference/chaptr01/wrlsnfrr/inftrans.htm

# THEORY

Mode settings of an air-conditioning unit

| MODE | FUNCTION | OPERATION | USED |
|---|---|---|---|
| Cool Mode | Cools room to desired temperature (16- 21°C) | Compressor: ON<br>Fan: ON | Hot days |
| Fan Mode | Circulates air | Compressor: OFF<br>Fan: ON | Ventilation/ Room already comfortable |
| Dry Mode | Removes humidity from the air | Compressor: ON in low speed | Humid days |
| Auto Mode | Automatic switches based on room temperature | Built-in sensors | Manage automatically |

# THEORY

## WORKING PRINCIPLES OF SENSORS

### Temperature Sensors

| Parameters | RTD | Thermocouple | Thermistor | Infrared |
|---|---|---|---|---|
| Materials | Platinum | Two dissimilar metals | Ceramic semiconductor | IR-Detecting images |
| Accuracy | Very high (±0.1°C–±0.5°C) | Moderate (±1°C–±3°C) | High (±0.1°C–±1°C) | Moderate - High |
| Temp Range | –200°C to +600°C | –200°C to +1750°C | –50°C to +150°C | –20°C to +2000°C |
| Response Time | 0.5 - 2 s | 0.1 - 5 s | 0.1 - 2 s | 10 ms – 1 s |

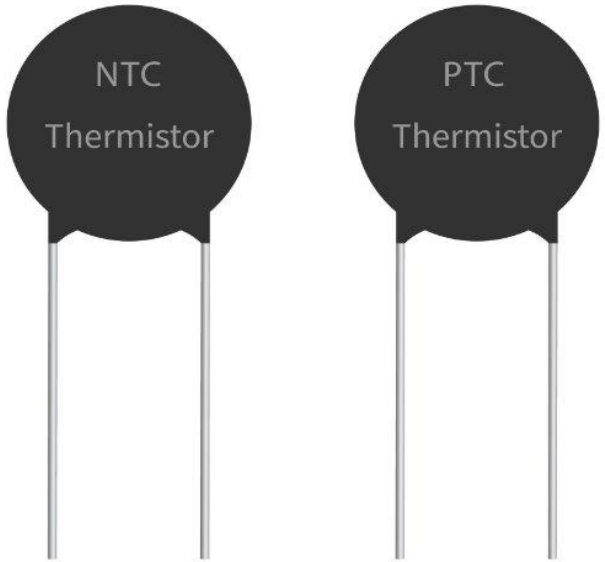
*https://www.sciencedirect.com/topics/earth-and-planetary-sciences/thermal-comfort*

# THEORY

## WORKING PRINCIPLES OF SENSORS

### Temperature Sensors

$$T = \frac{T_d}{16}$$




where:

$T$ = Temperature reading in $°C$

$T_d$ = binary digital output reading into decimal equivalent

*https://www.sciencedirect.com/topics/earth-and-planetary-sciences/thermal-comfort*

# THEORY

## WORKING PRINCIPLES OF SENSORS

### Humidity Sensors

| Parameters | Capacitive | Resistive | Thermal |
|---|---|---|---|
| Measurement Principles | Change in capacitance | Change in resistance | Thermal conductivity difference |
| Accuracy | High (±1°C–±3%) | Moderate (±2°C–±5%) | Low (±3°C–±10%) |
| Humidity Range | 0-100% RH | 5-95% RH | Limited |
| Response Time | 5-30 s | 5-60 s | 10-60 s |

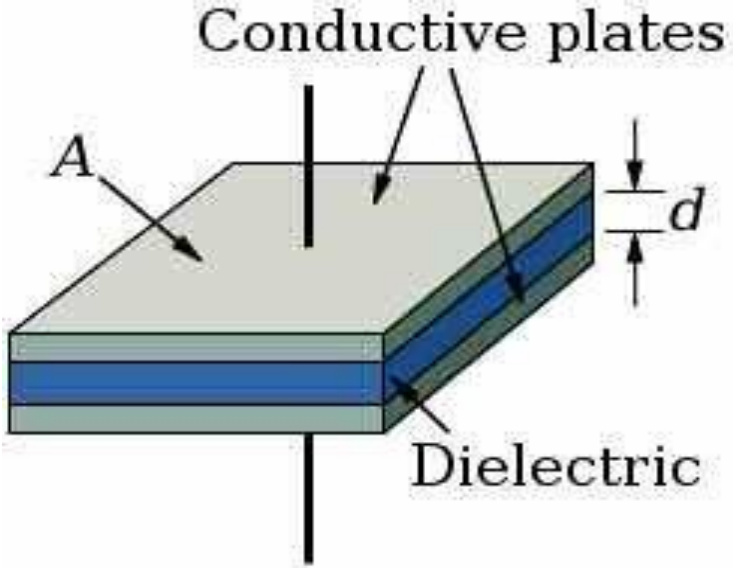
*https://www.sciencedirect.com/topics/earth-and-planetary-sciences/thermal-comfort*

# THEORY

## WORKING PRINCIPLES OF SENSORS

### Humidity Sensors

$$H = \frac{H_d}{16}$$




where:

$H$ = Humidity reading in %

$H_d$ = binary digital output reading into decimal equivalent

*https://www.sciencedirect.com/topics/earth-and-planetary-sciences/thermal-comfort*

# THEORY

## WORKING PRINCIPLES OF SENSORS

### Motion Sensors

| Parameters | PIR | Ultrasonic | Microwave |
|---|---|---|---|
| Detection Method | IR Heat Change | Ultrasonic wave | Microwave radar |
| Accuracy | 70-90% | 98-99% | 80-95% |
| Range | 5-12 m | 3-10 m | 5-30 m |
| Response Time | 0.5-1 s | 2-50 ms | 10-100 ms |

*https://www.sciencedirect.com/topics/earth-and-planetary-sciences/thermal-comfort*

# THEORY

## WORKING PRINCIPLES OF SENSORS

PIR current output

$$i = k \cdot \frac{dT}{dt}$$

where:

***i***= current generated

***k*** = sensitivity constant (50nA/°C)

***dT/dt*** = speed of temperature change hitting the sensor (W/m^2/s)

*https://www.sciencedirect.com/topics/earth-and-planetary-sciences/thermal-comfort*

# THEORY

## WORKING PRINCIPLES OF SENSORS

Current to voltage

$$V_{output} = i(R)$$

where:

$V_{output}$ = *Output voltage (v)*

$R$ = resistance

*https://www.sciencedirect.com/topics/earth-and-planetary-sciences/thermal-comfort*

# THEORY

## WORKING PRINCIPLES OF SENSORS

Detection method

$$V_{output} \geq V_{threshold}$$

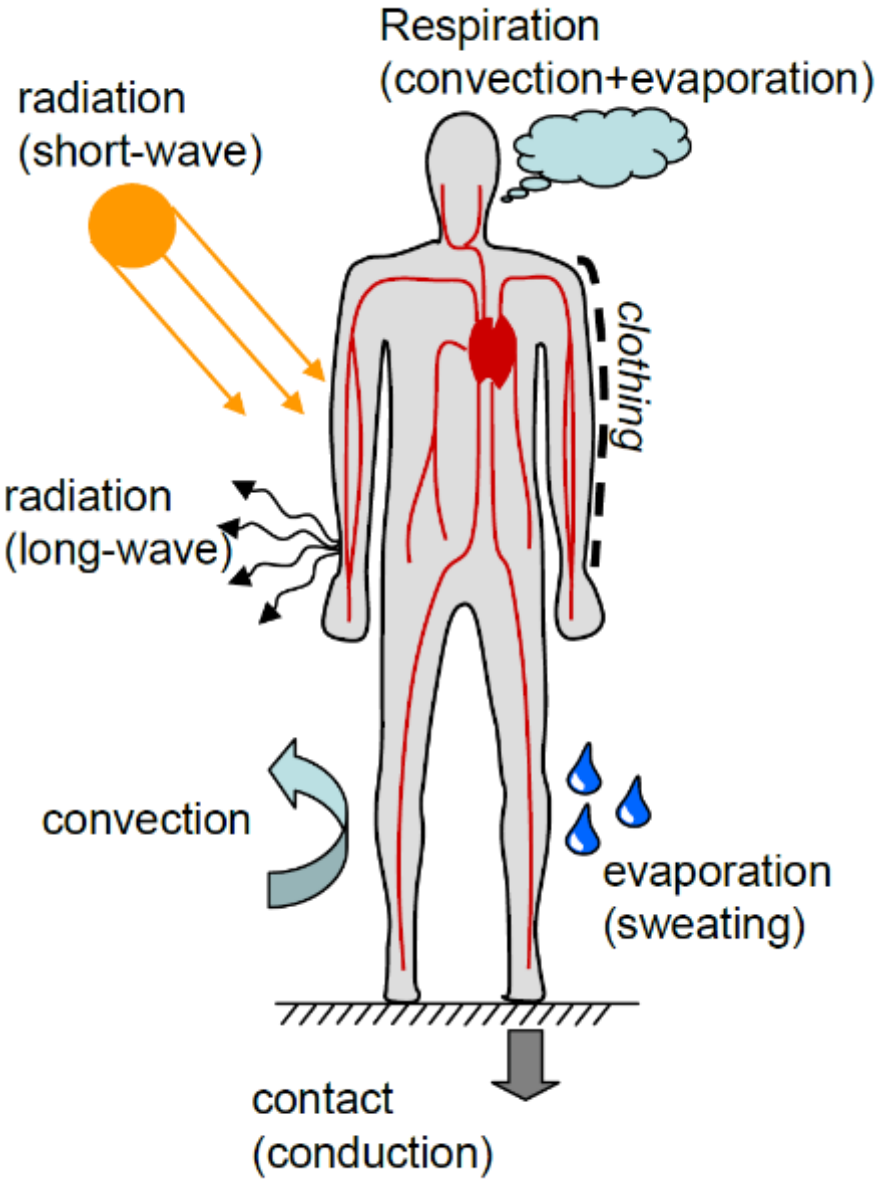
*https://www.sciencedirect.com/topics/earth-and-planetary-sciences/thermal-comfort*

# THEORY

## HUMAN THERMAL COMFORT

**Thermal Comfort**

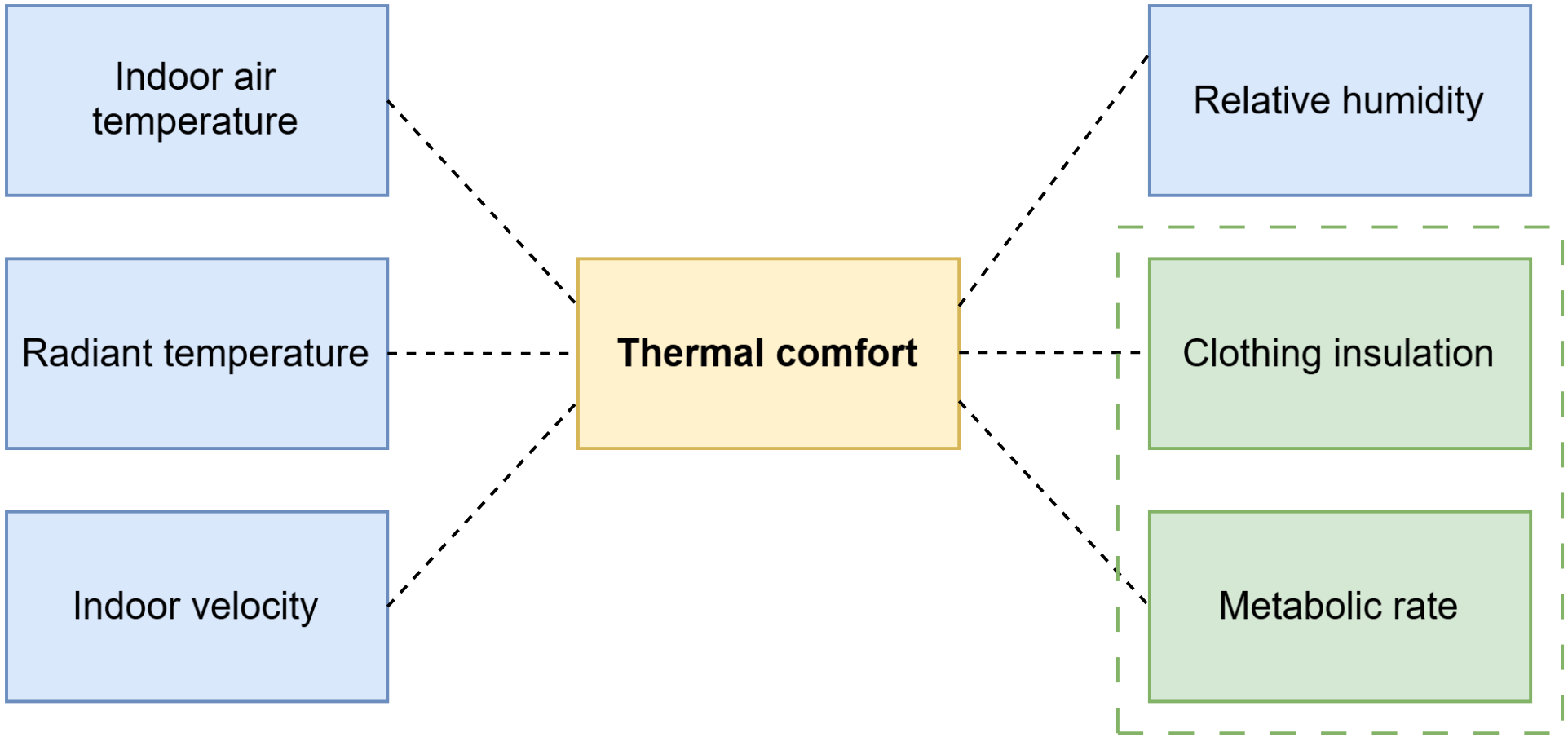
A *subjective* assessment and reflects satisfaction with the thermal environment




*https://www.sciencedirect.com/topics/earth-and-planetary-sciences/thermal-comfort*

# THEORY

## FACTORS AFFECTING THERMAL COMFORT




*https://www.sciencedirect.com/topics/earth-and-planetary-sciences/thermal-comfort*

# THEORY

## THERMAL COMFORT IN THE PHILIPPINES

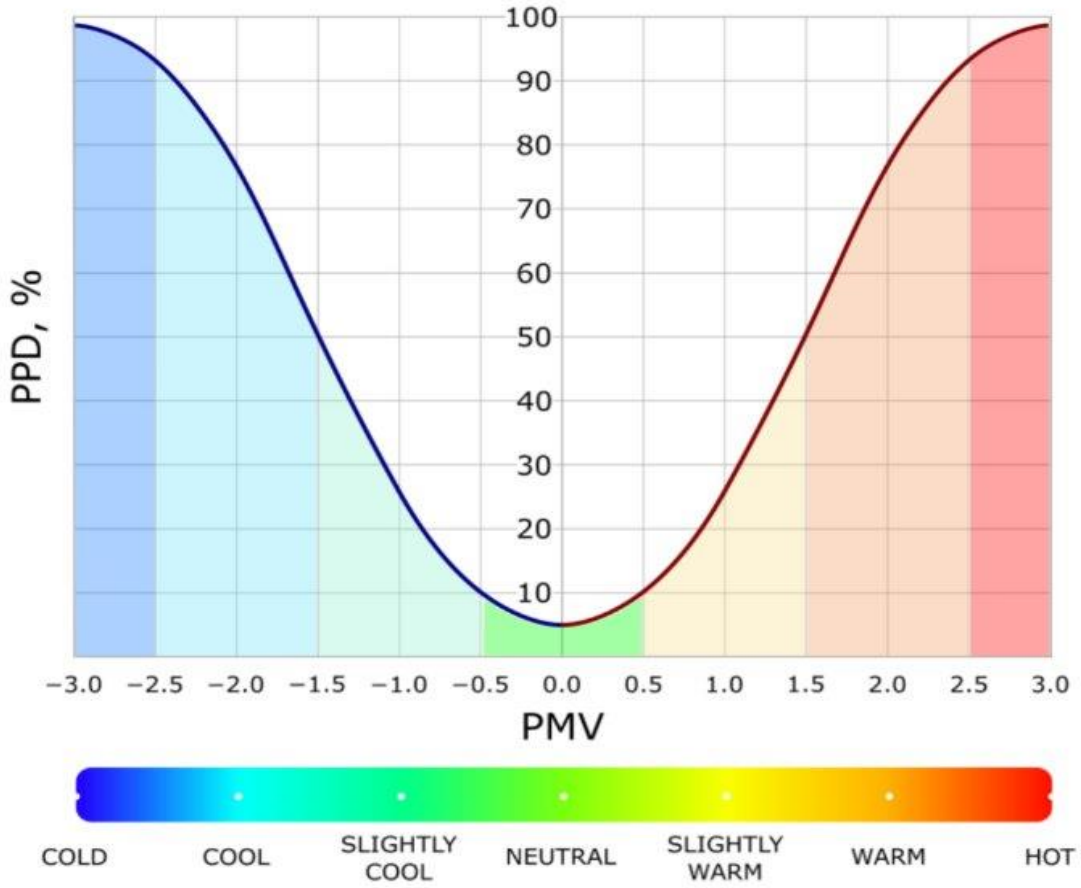
| Guidelines for Thermal Comfort | Desired Temp (°C) | Space Type |
|---|---|---|
| Enhancing Building Energy and Thermal Performance in the Tropics (ICFEE 2024) | 23 | Tropical |
| DOE: Government Energy Conservation Notice (April 2025) | 24 | Office & Classroom |
| DepEd No.168 s.2025: Direct Operational Guidance | 24-26 | Classroom |
| DOE "Guidelines on Energy Conserving Design of Buildings (2020)" | 24-26 | Buildings |

https://legacy.doe.gov.ph/press-releases/doe-urges-public-adopt-energy-efficiency-measures-summer-begins?
https://depedvigancity.com/wp-content/uploads/2024/06/Office-Memo-No.-024-s.-2024-Resumption-of-Conservation-and-Austerity-Measures-in-the-Use-of-Air-Conditioning-Units.pdf?
https://legacy.doe.gov.ph/sites/default/files/pdf/energy_efficiency/Building%20Guidelines%20User%20Manual_Final%20Pass.pdf?

# THEORY

## ANSI/ASHRAE STANDARD 55

Thermal environmental conditions for human occupancy




https://www.mdpi.com/1996-1073/15/6/2026

# THEORY

## ANSI/ASHRAE STANDARD 55

Seven-point thermal sensation scale

| Thermal Sensation | Vote |
|---|---|
| 3 | Hot |
| 2 | Warm |
| 1 | Slightly Warm |
| 0 | Neutral |
| -1 | Slightly Cool |
| -2 | Cool |
| -3 | Cold |

# THEORY

## ANSI/ASHRAE STANDARD 55

Predicted percentage of dissatisfied

| PMV | PPD | Comfort |
|---|---|---|
| 0 | 5% | Most people comfortable |
| ±0.5 | ~10% | Acceptable for office/classroom |
| ±1 | ~25% | Some discomfort, not ideal |
| ±2 | ~50% | Half of occupants dissatisfied |
| ±3 | ~100% | Almost everyone uncomfortable |

https://www.mdpi.com/1996-1073/15/6/2026

# THEORY

## ANSI/ASHRAE STANDARD 55

Thermal comfort requirements: ASHRAE 55

| Space Type | Air Temperature (°C) | Relative Humidity (%) | Air Speed (m/s) | Clothing insulation |
|---|---|---|---|---|
| Air-conditioned space | 23.5 - 25.5 | 70% | 0.15-0.25 | 0.5-1.0 |

https://www.mdpi.com/1996-1073/15/6/2026

# LITERATURE REVIEW

## MANUAL-REMOTE CONTROL SYSTEMS

Hardware:

| Common Features | Advantages | Disadvantages | Research Gap |
|---|---|---|---|
| • Operates via IR remote control<br>• Adjust mode functions manually<br>• Requires a line of sight between the remote and the aircon<br>• Works offline<br>• Usually brand-specific compatibility | • Low cost and easy to implement<br>• No internet communication<br>• Reliable in small spaces | • Requires constant human interactions<br>• No energy saving feature<br>• Not practical for large-scale areas | • Higher chance that the AC remains ON if the users forget to turn it OFF manually |

# REFERENCES


*[1] V. Ramani, M. Martin, P. Arjunan, A. Chong, K. Poolla and C. Miller, "Longitudinal thermal imagingfor scalable non-residential HVAC and occupant behavior characterization," doi: 10.3123/hvac/sui1762*

*[2] Z. Guo, X. Wang, Y. Wang, F. Zhu, H. Zhou, M. Zhang and Y. Wang, "The Cost-Optimal Control ofBuilding Air Conditioner Loads Based on Machine Learning: A Case Study of an Office Building in Nanjing," Buildings, vol. 14, no. 10, Art. 3040, Sep. 2024, doi: 10.3390/buildings14103040.*

*[3] J.-H. Ham, B.-S. Kim, I.-W. Bae and J. Joe, "Occupant-Detection-Based Individual Control of Four-Way Air Conditioner for Sustainable Building Energy Management," Sustainability, vol. 16, no.17,Art. 7404, Aug. 2024, doi: 10.3390/su16177404.*

*[4] A. M. Hashmi, I. A. Alkabiri, K. W. Alnajjar, H. H. Alattas, and T. H. Almutaiti, "Design of a Smart Air Conditioning Controller Based on the Occupancy of a Building," International Journal of Electrical and Electronic Engineering & Telecommunications, vol. 10, no. 3, pp. 160–167, May2021, doi:10.18178/ijeetc.10.3.160-167.*

# LITERATURE REVIEW

## AUTOMATIC-REMOTE CONTROL SYSTEMS

Website/Software:

| Common Features | Advantages | Disadvantages | Research Gap |
|---|---|---|---|
| • The system uses IoT technology for remote control of the AC<br>• Can control the increase and decrease of the temperature based on the user's preference<br>• Can automate the ON/OFF control of the aircon | • Minimizes manual operation and reduces human error<br>• Allows the users to control the AC remotely through IoT<br>• Convenient | • Dependent on stable connectivity<br>• Users have limited control over other remote settings of the AC | • No temperature monit oring<br>• No automatic mode and temperature a djustment |

# REFERENCES


*[1] M. A. Afandi, S. Nurandi and I. K. A. Enriko, "Automated Air Conditioner Controler and MonitoringBased on Internet of Things,"Indones. J. Electron. Instrum. Syst., vol. 11, no. 1, p. 83,Apr. 2021, doi: 10.22146/ijeis.64563.*

*[2] I. N. Sukarma, B. Anakottapary, I. G. K. S. Budarsa, and I. K. Parti, "Air Conditioner (AC) OperationUsing the Internet ofThings," J. Penelit. Pendidik. IPA, vol. 11, no. 4, pp. 575–582, Apr. 2025, doi:10.29303/jppipa.v11i4.10943.*

*[3] A. Maulana, R. D. W. and D. Setiawan, "Design and Build Smart Remote for IoT Based Air Conditioners," J. Impresi Indonesia, vol. 3, no. 11, Nov. 2024, doi: 10.58344/jii.v3i11.5711.*

*[4] X. Cui, V. G. Zakka and M. Lee, "A Computer Vision-Based Model for Occupancy Detection Using Low-Resolution Thermal Images," arXiv, May 13 2025.*

# LITERATURE REVIEW

## AUTOMATIC-REMOTE CONTROL SYSTEMS

Application:

| Common Features | Advantages | Disadvantages | Research Gap |
|---|---|---|---|
| • The system uses the Blynk Application to send temperature adjustment commands to the AC<br>• Can automate the ON/OFF control of the aircon<br>• Support app-based control | • Enable to control the aircon anywhere if there is an internet connection<br>• User-friendly<br>• Convenient | • Dependent on stable connectivity<br>• No temperature monitoring<br>• Does not support other AC mode settings | • No temperature monit oring<br>• No control over other mode settings of the aircon |

# REFERENCES


*[5] X. Cui, V. G. Zakka and M. Lee, "A Computer Vision-Based Model for Occupancy Detection Using Low-Resolution Thermal Images," arXiv, May 13 2025.*

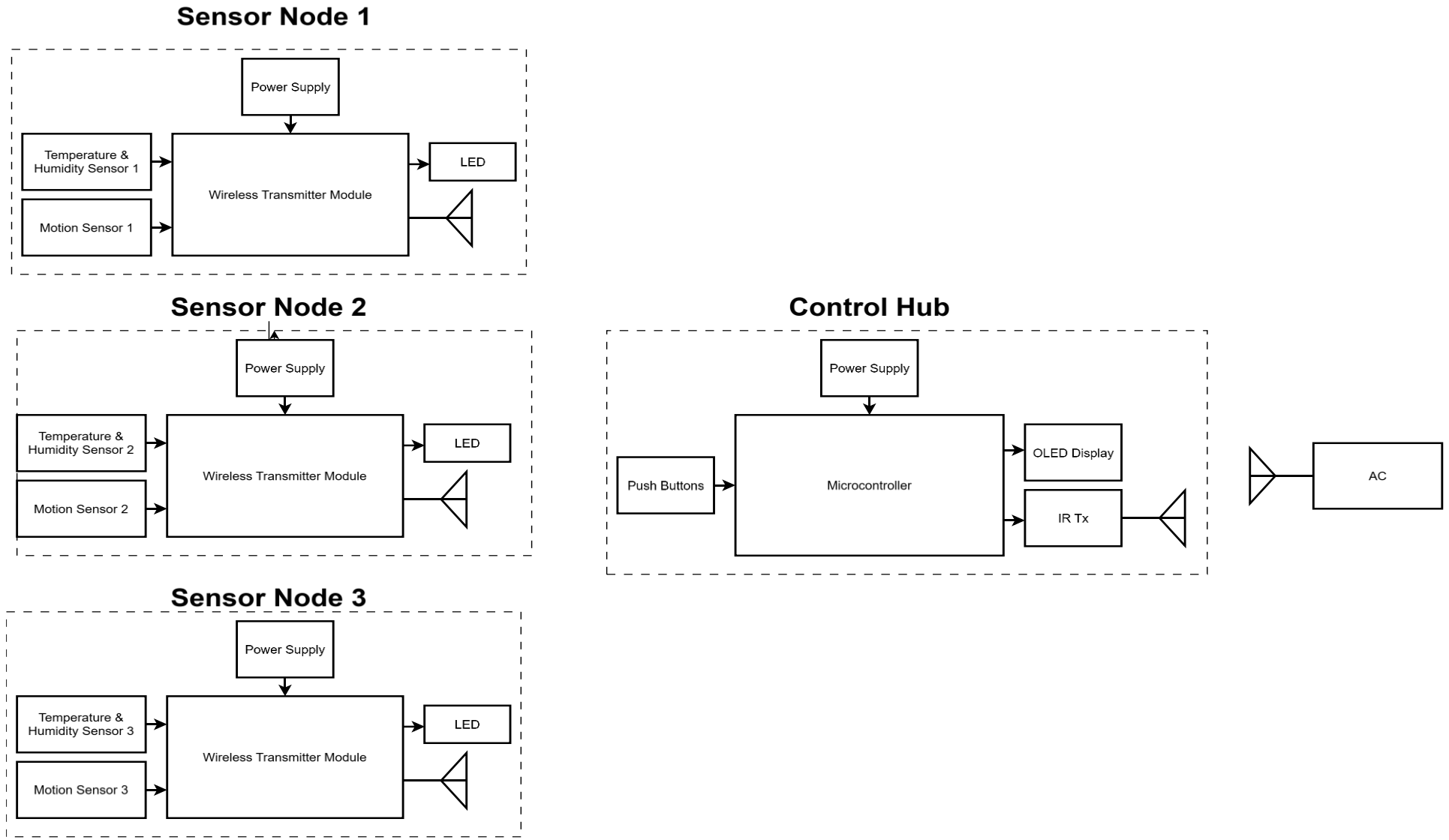
*[6] B. Zhang, "Research on Intelligent Control System of Air Conditioning Based on Internet of Things," International Journal of Advanced Computer Science and Applications (IJACSA), vol. 13, no. 9, pp. 803–814, 2022.*

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*[8] N. H. A. Shukor, Z. M. P. A. Baidowi, M. R. M. Isa, M. Y. Darus, and M. Abdullah, "Switching On/OffAir Conditioner and Fan Alternately based on IoT Motion Detection and Room Temperature, "International Journal on Informatics Visualization, vol. 8, no. 3-2, pp. 1560–1566, Nov. 2024.*

# METHODOLOGY

## SYSTEM BLOCK DIAGRAM



# DECISION TABLE

## CASE 1: ALL NODES ARE EQUAL IN TEMPERATURE

| Sensor node 1 | Sensor node 2 | Sensor node 3 | Output |
|---|---|---|---|
| ≤16°C | ≤16°C | ≤16°C | Mode = Fan<br>Temp = 32<br>Swing = Off |
| 18-19°C | 18-19°C | 18-19°C | Mode = Fan<br>Temp = 30<br>Swing = Off |
| 20-21°C | 20-2°C | 20-21°C | Mode = Fan<br>Temp = 28<br>Swing = Off |
| 22-23°C | 22-23°C | 22-23°C | Mode = Fan<br>Temp = 26<br>Swing = Off |
| 24-24.9°C | 24-24.9°C | 24-24.9°C | Mode = Fan<br>Temp = 24<br>Swing = Off |
| 25-25.9°C | 25-25.9°C | 25-25.9°C | Mode = Cool<br>Temp = 24<br>Swing = Auto |
| | | | Continue to next page |

# DECISION TABLE

## CASE 1: ALL NODES ARE EQUAL IN TEMPERATURE

| Sensor node 1 | Sensor node 2 | Sensor node 3 | Output |
|---|---|---|---|
| 26-26.9°C | 26-26.9°C | 26-26.9°C | Mode = Cool<br>Temp = 23<br>Swing = Auto |
| 27-28°C | 27-28°C | 27-28°C | Mode = Cool<br>Temp = 22<br>Swing = Auto |
| 29-30°C | 29-30°C | 29-30°C | Mode = Cool<br>Temp = 20<br>Swing = Auto |
| ≥31°C | ≥31°C | ≥31°C | Mode = Cool<br>Temp = 16<br>Swing = Auto |

# TRUTH TABLE: Boolean Expression

## CASE 1: ALL NODES ARE EQUAL IN TEMPERATURE

| SN1 | SN2 | SN3 | Output |
|---|---|---|---|
| 0 | 0 | 0 | Mode = Fan<br>Temp = 32<br>Swing = Off |
| 0 | 0 | 1 | Mode = Fan<br>Temp = 30<br>Swing = Off |
| 0 | 1 | 0 | Mode = Fan<br>Temp = 28<br>Swing = Off |
| 0 | 1 | 1 | Mode = Fan<br>Temp = 26<br>Swing = Off |
| 1 | 0 | 0 | Mode = Fan<br>Temp = 24<br>Swing = Off |
| 1 | 0 | 1 | Mode = Cool<br>Temp = 24<br>Swing = Auto |
| Continue... | | | |

| SN1 | SN2 | SN3 | Output |
|---|---|---|---|
| 1 | 1 | 0 | Mode = Cool<br>Temp = 23<br>Swing = Auto |
| 1 | 1 | 1 | Mode = Cool<br>Temp = 22<br>Swing = Auto |
| x | 0 | 0 | Mode = Cool<br>Temp = 20<br>Swing = Auto |
| x | 0 | 1 | Mode = Cool<br>Temp = 16<br>Swing = Auto |

# DECISION TABLE

## CASE 2: HOTTEST NODE

| Sensor node 1 | Sensor node 2 | Sensor node 3 | Output |
|---|---|---|---|
| 16-16.9°C | ≤15°C | ≤15°C | Mode = Fan<br>Temp = 32<br>Swing = Off |
| 17-17.9°C | ≤16°C | ≤16°C | Mode = Fan<br>Temp = 31<br>Swing = Off |
| 18-18.9°C | ≤17°C | ≤17°C | Mode = Fan<br>Temp = 30<br>Swing = Off |
| 19-19.9°C | ≤18°C | ≤18°C | Mode = Fan<br>Temp = 29<br>Swing = Off |
| 20-20.9°C | ≤19°C | ≤19°C | Mode = Fan<br>Temp = 28<br>Swing = Off |
| 21-21.9°C | ≤20°C | ≤20°C | Mode = Fan<br>Temp = 27<br>Swing = Off |

Continue to next page

# DECISION TABLE

## CASE 2: HOTTEST NODE

| Sensor node 1 | Sensor node 2 | Sensor node 3 | Output |
|---|---|---|---|
| 22-22.9°C | ≤21°C | ≤21°C | Mode = Fan<br>Temp = 26<br>Swing = Off |
| 23-23.9°C | ≤22°C | ≤22°C | Mode = Fan<br>Temp = 25<br>Swing = Off |
| 24-24.9°C | ≤23°C | ≤23°C | Mode = Fan<br>Temp = 25<br>Swing = Auto |
| 25-25.9°C | ≤24°C | ≤24°C | Mode = Cool<br>Temp = 24<br>Swing = On (Focus on hottest_node) |
| 26-26.9°C | ≤25°C | ≤25°C | Mode = Cool<br>Temp = 23<br>Swing = On (Focus on hottest_node) |

Continue to next page

# DECISION TABLE

## CASE 2: HOTTEST NODE

| Sensor node 1 | Sensor node 2 | Sensor node 3 | Output |
|---|---|---|---|
| 27-27.9°C | ≤26°C | ≤26°C | Mode = Cool<br>Temp = 22<br>Swing = On (Focus on hottest_node) |
| 28-28.9°C | ≤27°C | ≤27°C | Mode = Cool<br>Temp = 21<br>Swing = On (Focus on hottest_node) |
| 29-29.9°C | ≤28°C | ≤28°C | Mode = Cool<br>Temp = 20<br>Swing = On (Focus on hottest_node) |
| 30-30.9°C | ≤29°C | ≤29°C | Mode = Cool<br>Temp = 19<br>Swing = On (Focus on hottest_node) |
| 31-31.9°C | ≤30°C | ≤30°C | Mode = Cool<br>Temp = 23<br>Swing = On (Focus on hottest_node) |
| ≥32 | ≤31°C | ≤31°C | Mode = Cool<br>Temp = 16<br>Swing = On (Focus on hottest_node) |

# TRUTH TABLE: Boolean Expression

## CASE 2: HOTTEST NODE

| SN1 | SN2 | SN3 | Output |
|---|---|---|---|
| 0 | 0 | 0 | Mode = Fan<br>Temp = 32<br>Swing = Off |
| 0 | 0 | 1 | Mode = Fan<br>Temp = 31<br>Swing = Off |
| 0 | 1 | 0 | Mode = Fan<br>Temp = 30<br>Swing = Off |
| 0 | 1 | 1 | Mode = Fan<br>Temp = 29<br>Swing = Off |
| 1 | 0 | 0 | Mode = Fan<br>Temp = 28<br>Swing = Off |
| 1 | 0 | 1 | Mode = Fan<br>Temp = 27<br>Swing = Off |
| | | | Continue... |

| SN1 | SN2 | SN3 | Output |
|---|---|---|---|
| 1 | 1 | 0 | Mode = Fan<br>Temp = 26<br>Swing = Off |
| 1 | 1 | 1 | Mode = Fan<br>Temp = 25<br>Swing = Off |
| x | 0 | 0 | Mode = Fan<br>Temp = 25<br>Swing = Auto |
| x | 0 | 1 | Mode = Cool<br>Temp = 24<br>Swing = On (Focus on hottest_node) |
| x | 1 | 0 | Mode = Cool<br>Temp = 23<br>Swing = On (Focus on hottest_node) |
| x | 1 | 1 | Mode = Cool<br>Temp = 22<br>Swing = On (Focus on hottest_node) |
| | | | Continue to next page |

# TRUTH TABLE: Boolean Expression

## CASE 2: HOTTEST NODE

| SN1 | SN2 | SN3 | Output |
|---|---|---|---|
| 0 | x | 0 | Mode = Cool<br>Temp = 21<br>Swing = On (Focus on hottest_node) |
| 0 | x | 1 | Mode = Cool<br>Temp = 20<br>Swing = On (Focus on hottest_node) |
| 1 | x | 0 | Mode = Cool<br>Temp = 19<br>Swing = On (Focus on hottest_node) |
| 1 | x | 1 | Mode = Cool<br>Temp = 18<br>Swing = On (Focus on hottest_node) |
| x | x | x | Mode = Cool<br>Temp = 16<br>Swing = On (Focus on hottest_node) |

# DECISION TABLE

## CASE 3: 2 HOT NODES ARE EQUAL

| Sensor node 1 | Sensor node 2 | Sensor node 3 | Output |
|---|---|---|---|
| 19°C | 19°C | ≤18.9°C | Mode = Fan<br>Temp = 30<br>Swing = Off |
| 20°C | 20°C | ≤19°C | Mode = Fan<br>Temp = 28<br>Swing = Off |
| 21°C | 21°C | ≤20°C | Mode = Fan<br>Temp = 27<br>Swing = Off |
| 22°C | 22°C | ≤21°C | Mode = Fan<br>Temp = 25<br>Swing = Off |
| 23°C | 23°C | ≤22°C | Mode = Fan<br>Temp = 25<br>Swing = Auto |
| 24-25°C | 25-25.9°C | ≤23°C | Mode = Cool<br>Temp = 24<br>Swing = Auto |
| | | | Continue to next page |

# DECISION TABLE

## CASE 3: 2 HOT NODES ARE EQUAL

| Sensor node 1 | Sensor node 2 | Sensor node 3 | Output |
|---|---|---|---|
| 26°C | 26°C | ≤25°C | Mode = Cool<br>Temp = 22<br>Swing = Auto |
| 27°C | 27°C | ≤26°C | Mode = Cool<br>Temp = 21<br>Swing = Auto |
| 28°C | 28°C | ≤27°C | Mode = Cool<br>Temp = 20<br>Swing = Auto |
| 29°C | 29°C | ≤28°C | Mode = Cool<br>Temp = 19<br>Swing = Auto |
| 30°C | 30°C | ≤29°C | Mode = Cool<br>Temp = 18<br>Swing = Auto |
| ≥31°C | ≥31°C | ≤30°C | Mode = Cool<br>Temp = 16<br>Swing = Auto |

# TRUTH TABLE: Boolean Expression

## CASE 3: 2 HOT NODES ARE EQUAL

| SN1 | SN2 | SN3 | Output |
|---|---|---|---|
| 0 | 0 | 0 | Mode = Fan<br>Temp = 30<br>Swing = Off |
| 0 | 0 | 1 | Mode = Fan<br>Temp = 28<br>Swing = Off |
| 0 | 1 | 0 | Mode = Fan<br>Temp = 27<br>Swing = Off |
| 0 | 1 | 1 | Mode = Fan<br>Temp = 25<br>Swing = Off |
| 1 | 0 | 0 | Mode = Fan<br>Temp = 25<br>Swing = Auto |
| 1 | 0 | 1 | Mode = Cool<br>Temp = 24<br>Swing = Auto |
| Continue... | | | |

| SN1 | SN2 | SN3 | Output |
|---|---|---|---|
| 1 | 1 | 0 | Mode = Cool<br>Temp = 22<br>Swing = Auto |
| 1 | 1 | 1 | Mode = Cool<br>Temp = 21<br>Swing = Auto |
| x | 0 | 0 | Mode = Cool<br>Temp = 20<br>Swing = Auto |
| x | 0 | 1 | Mode = Cool<br>Temp = 19<br>Swing = Auto |
| x | 1 | 0 | Mode = Cool<br>Temp = 18<br>Swing = Auto |
| x | 1 | 1 | Mode = Cool<br>Temp = 16<br>Swing = Auto |

# DECISION TABLE

## CASE 4: HUMIDITY

| Sensor node 1 | Sensor node 2 | Sensor node 3 | High_nodes | Output |
|---|---|---|---|---|
| ≥70% | ≥70% | ≥70% | 3 | Mode = Dry<br>Temp = 24<br>Swing = Auto |
| ≥70% | ≥70% | ≥69% | 2 | Mode = Dry<br>Temp = 24<br>Swing = Auto |
| ≥70% | ≥69% | ≥69% | 1 | Mode = Cool<br>Temp = 24<br>Swing = Auto |
| ≥69% | ≥69% | ≥69% | 0 | Mode = Cool<br>Temp = 24<br>Swing = Off |

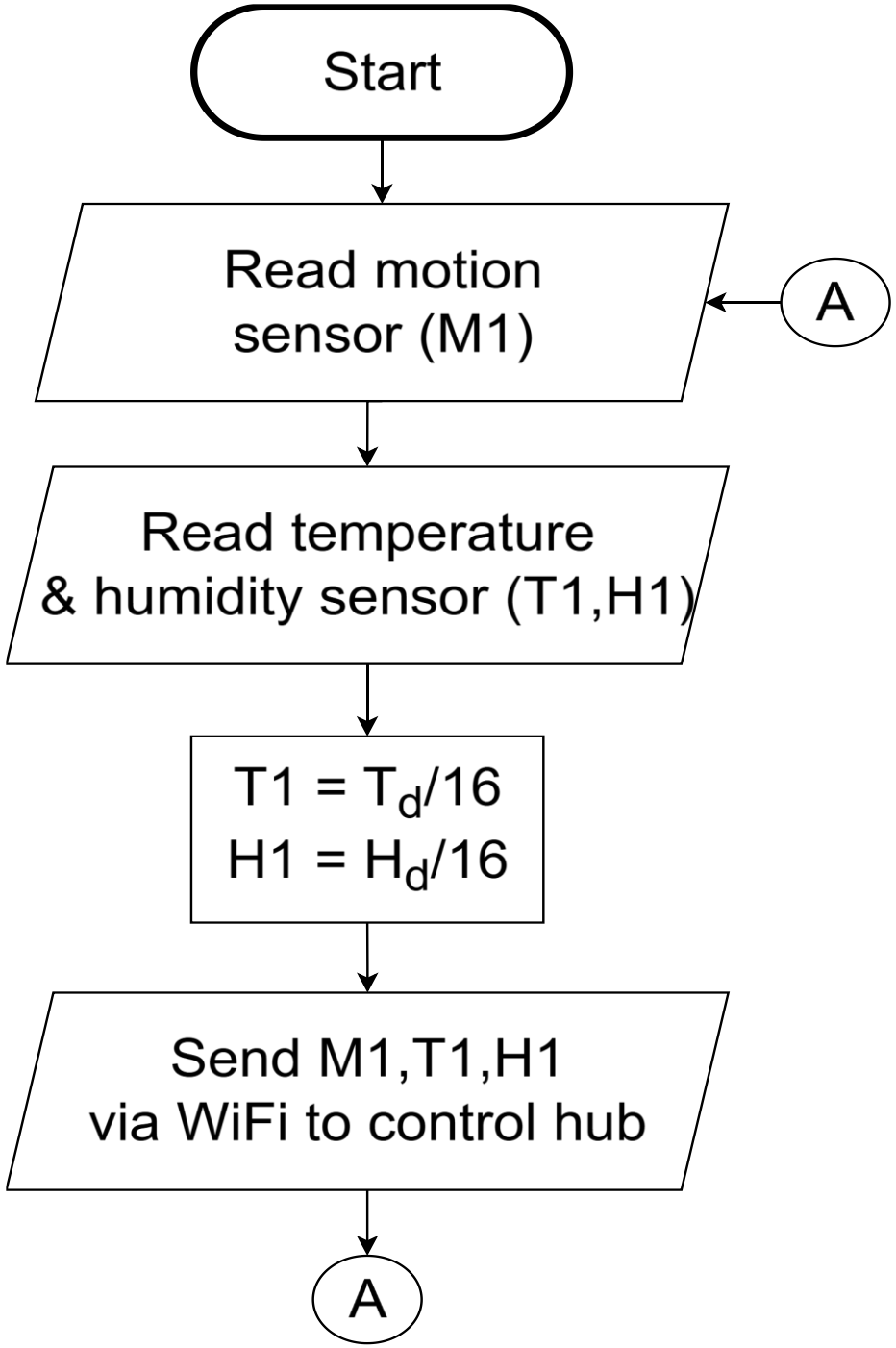
# TRUTH TABLE: Boolean Expression

## CASE 4: HUMIDITY

| SN1 | SN2 | SN3 | Output |
|---|---|---|---|
| 0 | 0 | 0 | Mode = Dry<br>Temp = 24<br>Swing = Auto |
| 0 | 0 | 1 | Mode = Dry<br>Temp = 24<br>Swing = Auto |
| 0 | 1 | 0 | Mode = Cool<br>Temp = 24<br>Swing = Auto |
| 0 | 1 | 1 | Mode = Cool<br>Temp = 24<br>Swing = Off |

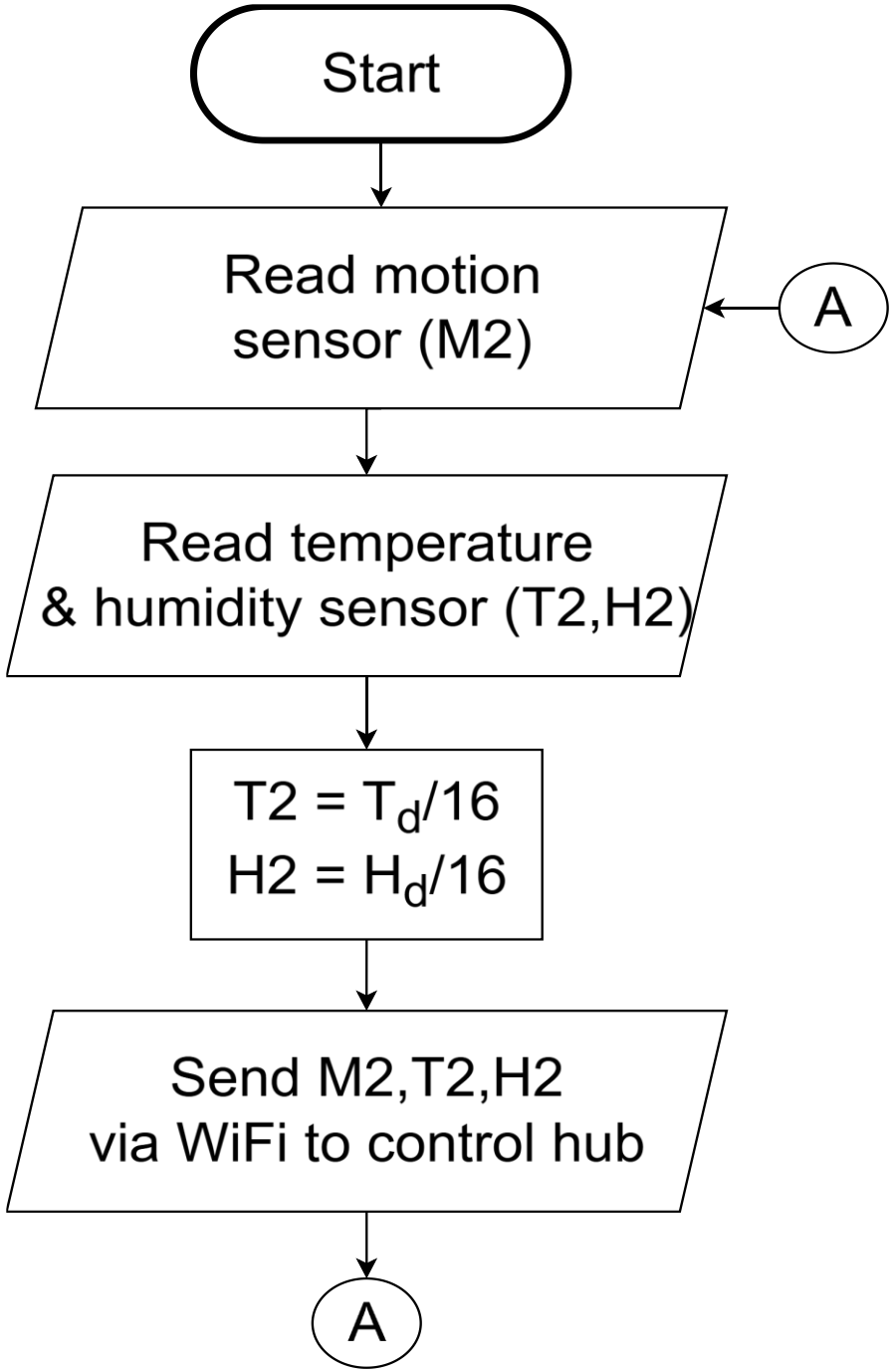
# METHODOLOGY

## SENSOR NODE 1



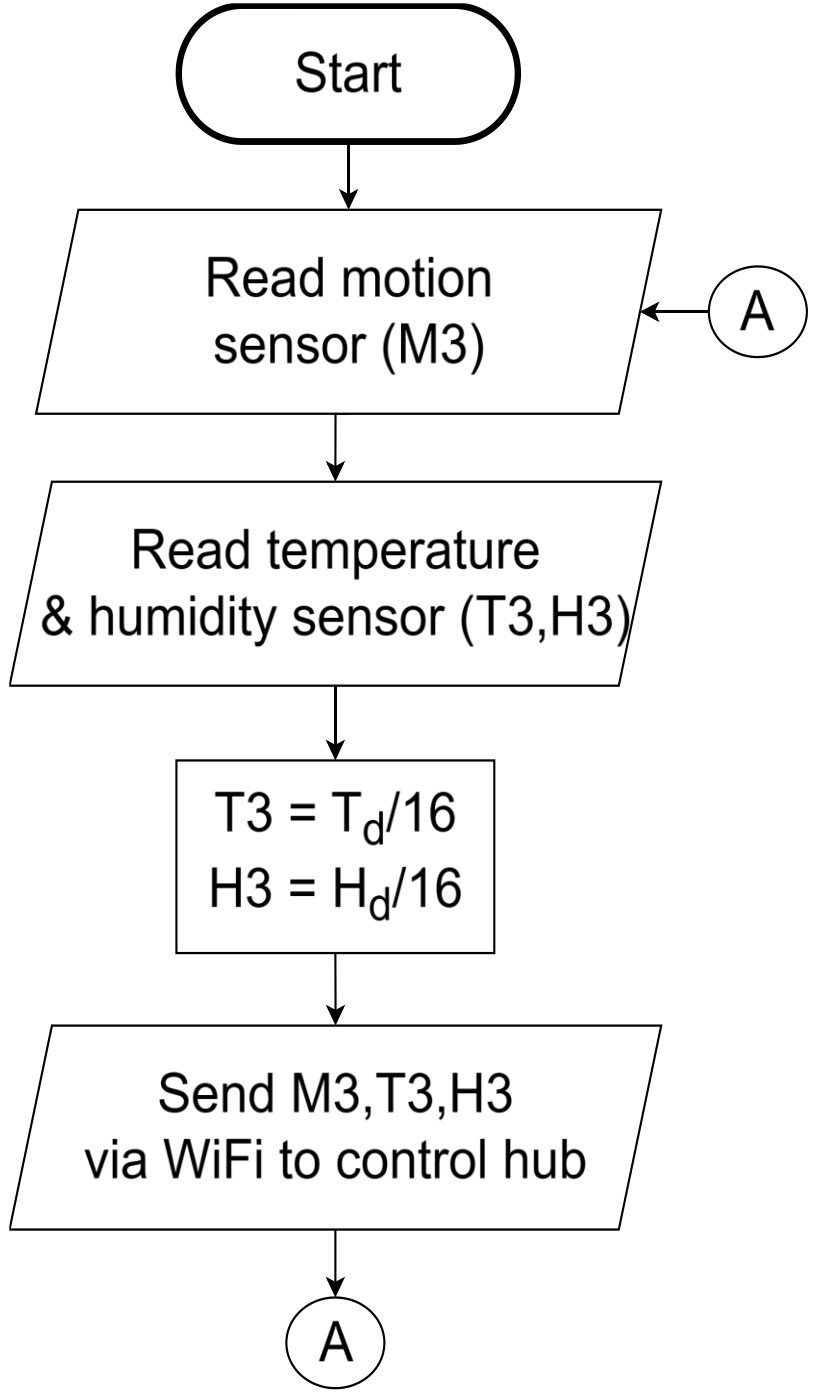
# METHODOLOGY

## SENSOR NODE 2



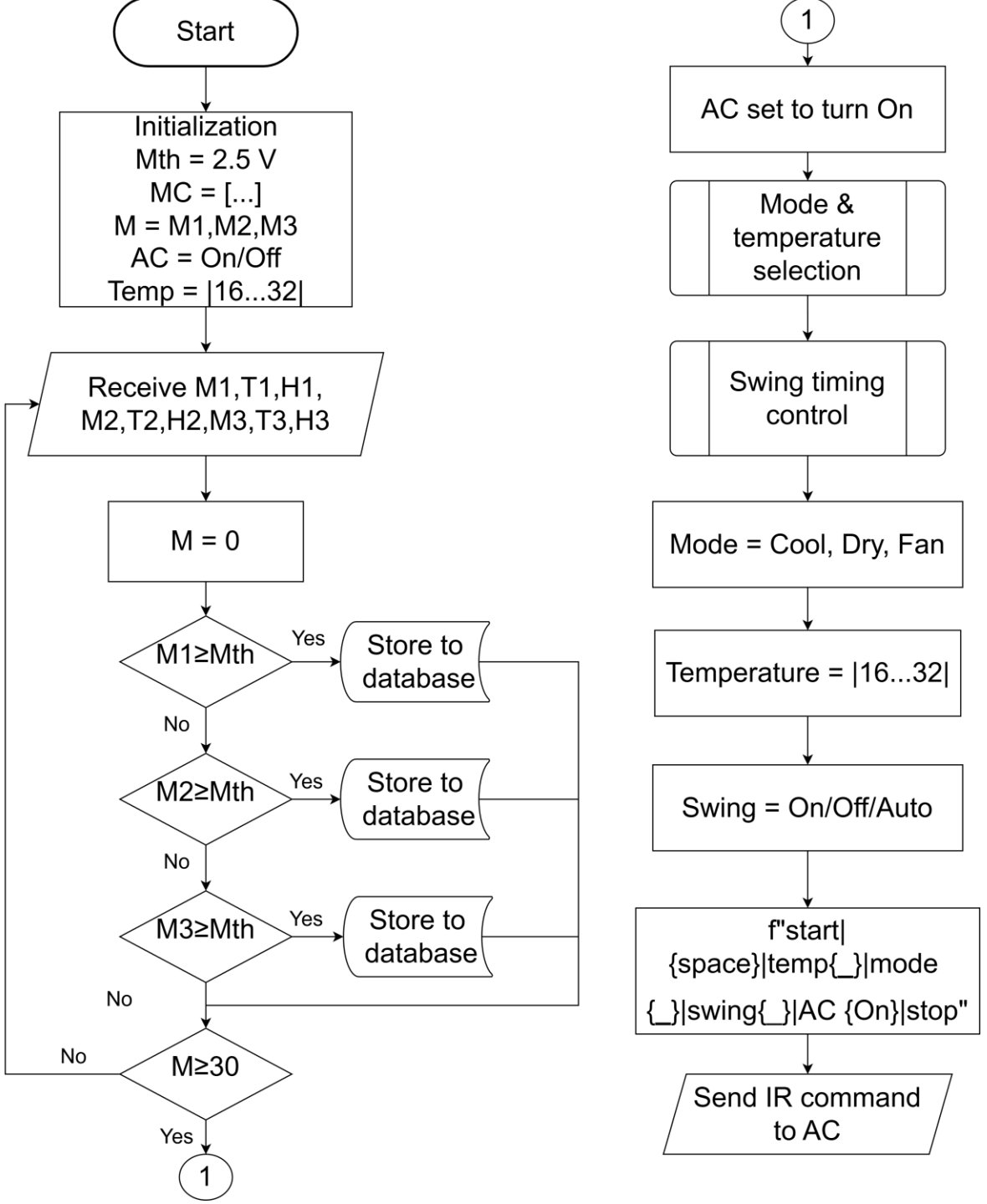
# METHODOLOGY

## SENSOR NODE 3

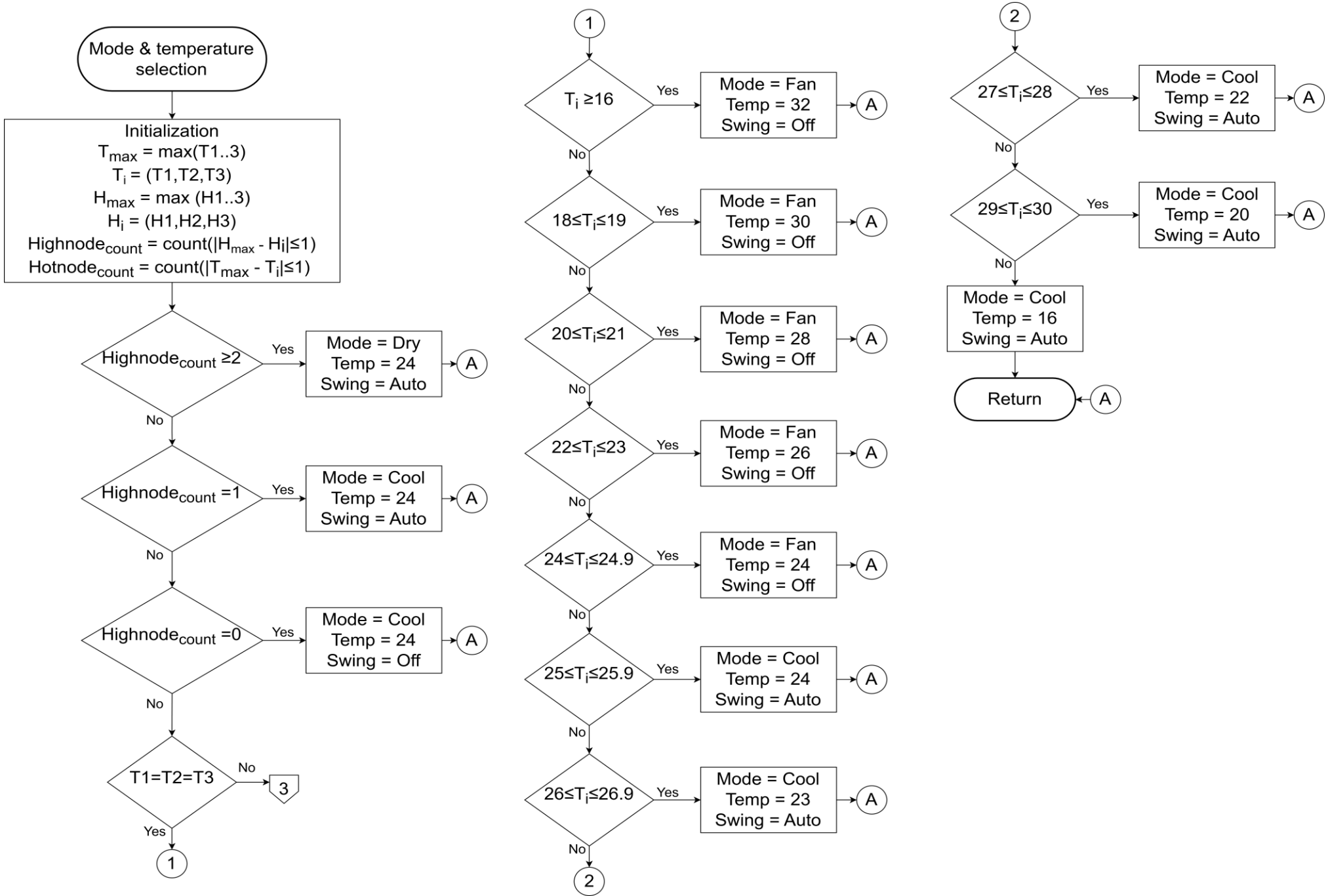


# METHODOLOGY

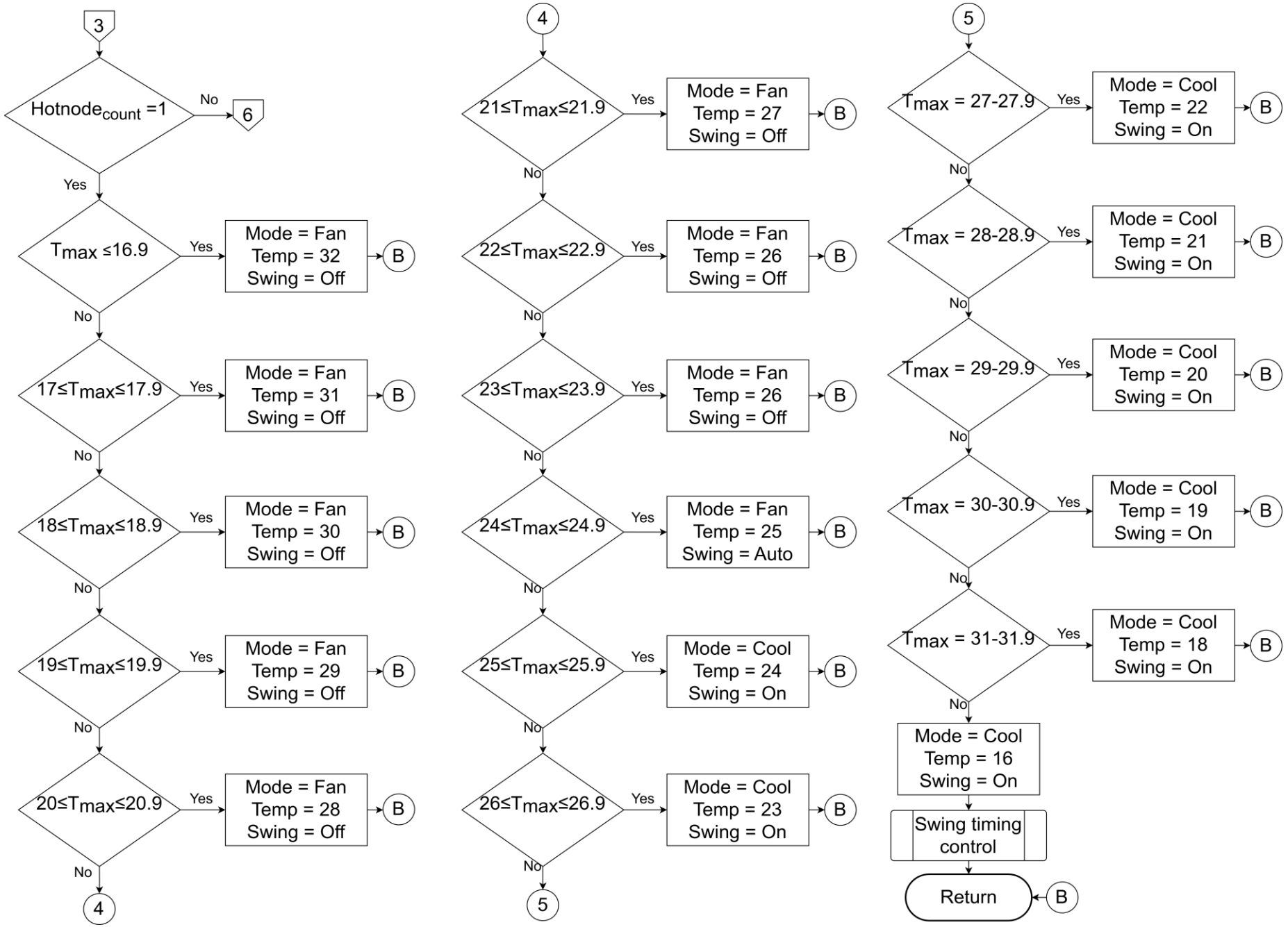
## CONTROL HUB



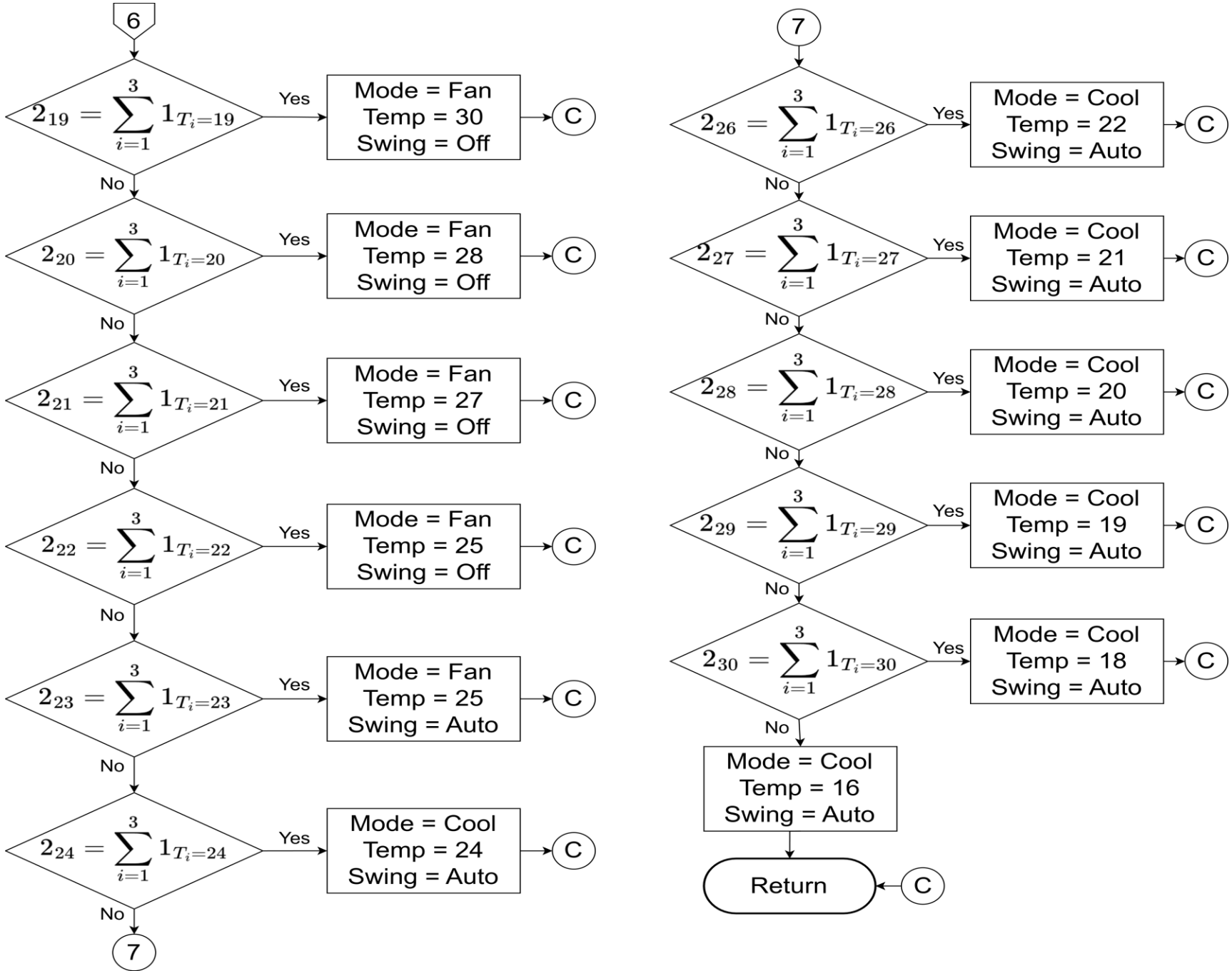
# SUBRPROCESS MODE & TEMPERATURE SELECTION



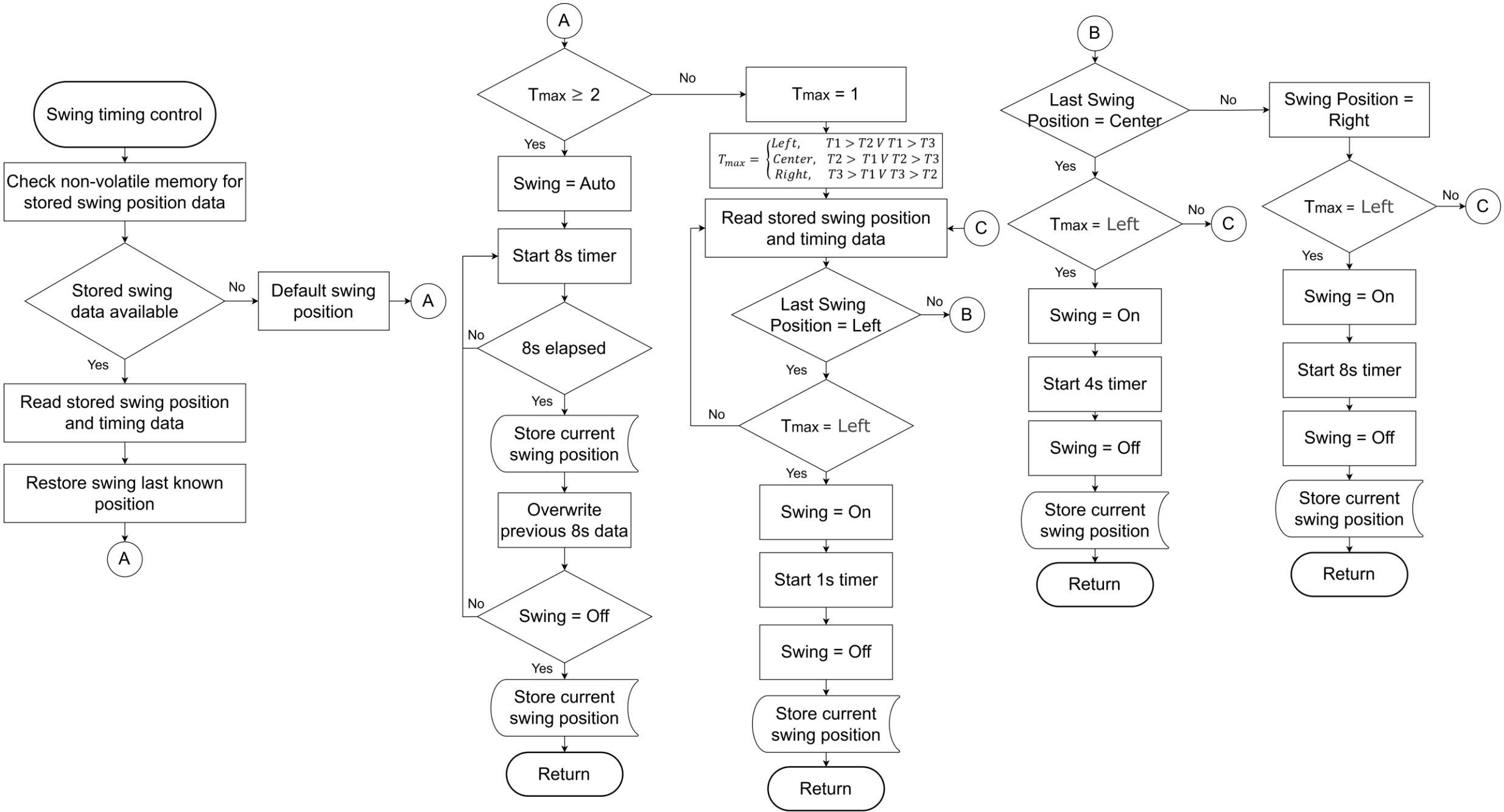
# SUBRPROCESS MODE & TEMPERATURE SELECTION



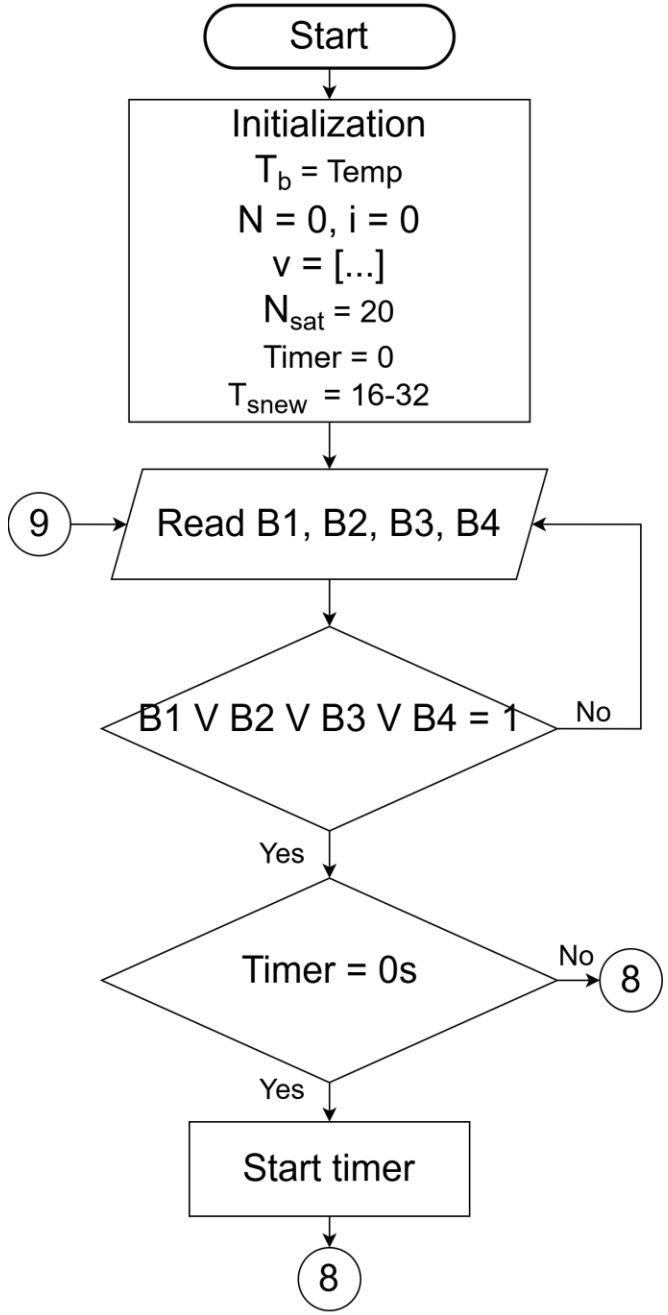
# SUBRPROCESS MODE & TEMPERATURE SELECTION

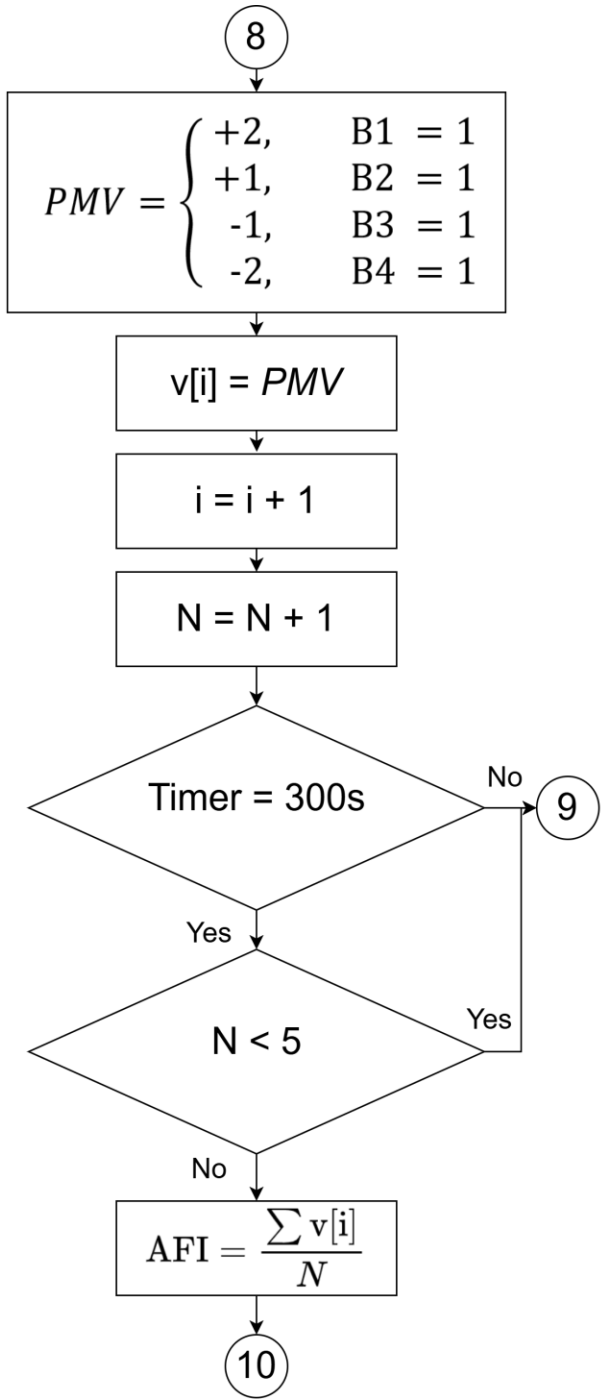


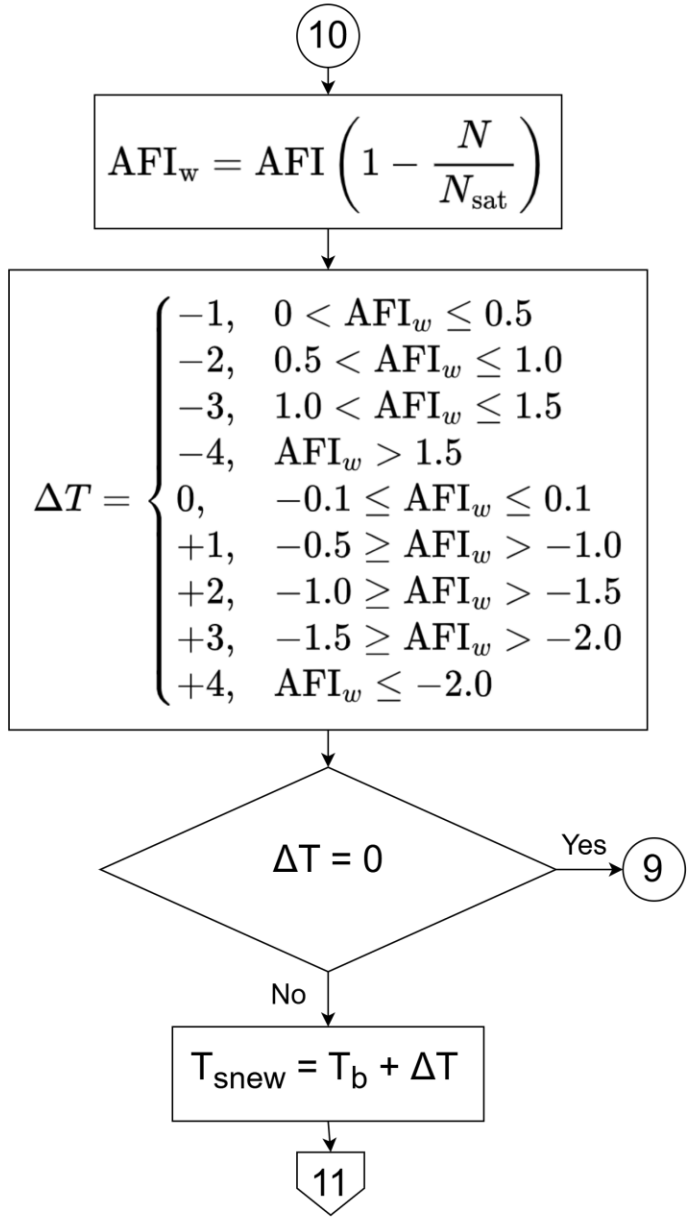
# SUBRPROCESS SWING TIMING CONTROL



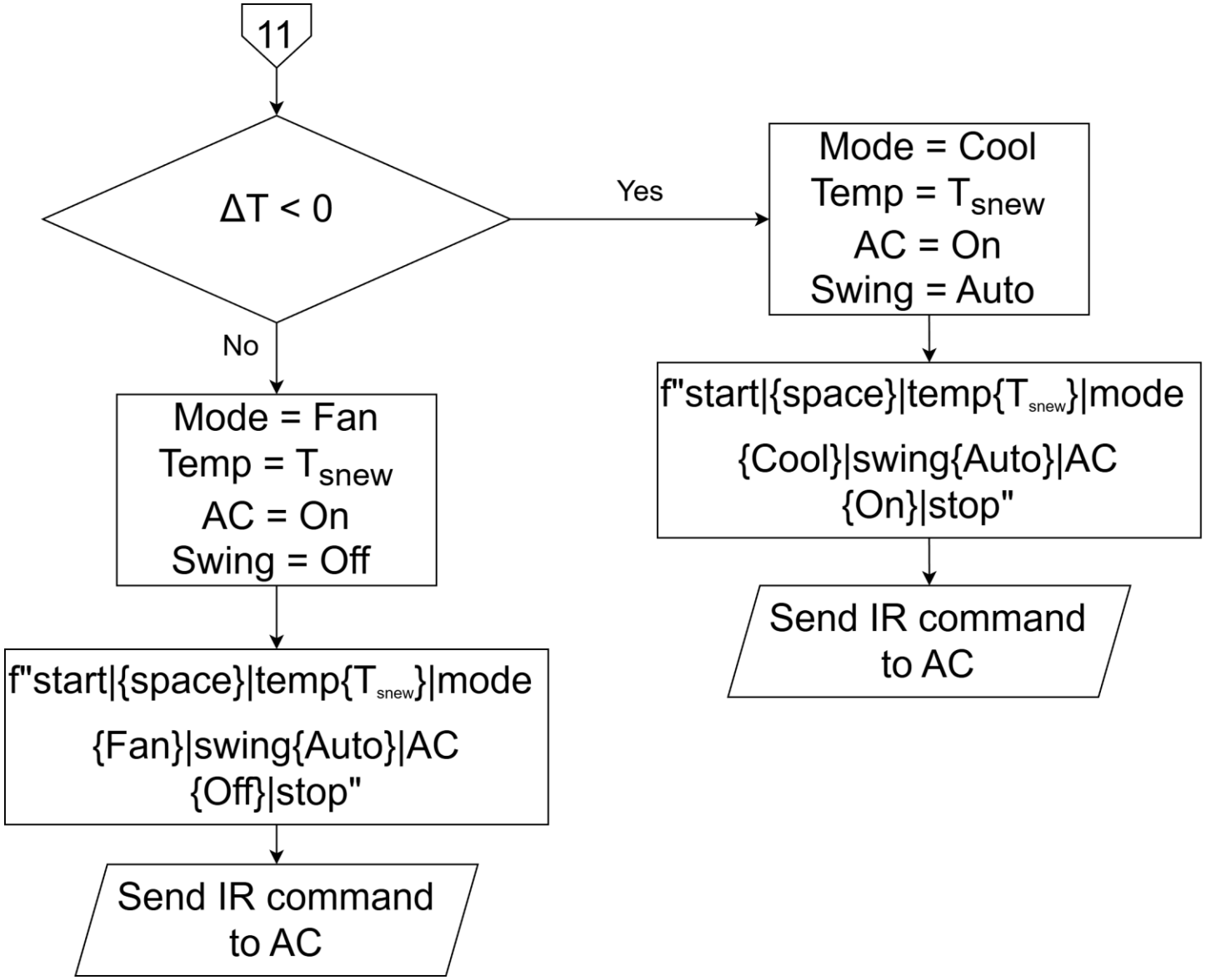
# VOTING MECHANISM



# VOTING MECHANISM



# METHODOLOGY

Assign **PMV** based on button pressed

| Button ID | Votes | PMV Scale Value |
|---|---|---|
| B1 | Hot | +2 |
| B2 | Slightly Warm | +1 |
| B3 | Slightly Cool | -1 |
| B4 | Cold | -2 |

# METHODOLOGY

Map AFIw to delta ΔT

| AFI weighted range | ΔT (°C) |
|---|---|
| $0 < AFIw \leq 0.5$ | -1 |
| $0.5 < AFIw \leq 1.0$ | -2 |
| $1.0 < AFIw \leq 1.5$ | -3 |
| $AFIw > 1.5$ | -4 |
| $-0.1 \leq AFIw \leq 0.1$ | 0 |
| $-0.5 \geq AFIw > -1.0$ | +1 |
| $-1.0 \geq AFIw > -1.5$ | +2 |
| $-1.5 \geq AFIw > -2.0$ | +3 |
| $AFIw \leq -2.0$ | +4 |

# METHODOLOGY

## HARDWARE COMPONENTS

### Temperature & Humidity Sensor

| Model Name/ No. | DHT22 °C |
|---|---|
| Measurement range: | 20-90% RH; 0-50 °C |
| Temperature accuracy: | ±2°C |
| Humidity accuracy: | ±5% |
| Operating voltage: | 3.6 V – 5 V |




### Purpose:

*Measures the room temperature and the humidity*

# METHODOLOGY

## HARDWARE COMPONENTS

### Infrared Transmitter

| Model Name/ No. | 940 nm IR LED |
|---|---|
| Peak wavelength: | 940 nm |
| Forward voltage: | 1.2 - 1.4 V |
| Max forward current: | 100 – 200 mA |
| Viewing angle: | 30 - degrees |



### Purpose:

*Sends encoded IR signals to the aircon*

# METHODOLOGY

## HARDWARE COMPONENTS

### PIR Motion Sensor

| Model Name/ No. | HCSR501 PIR |
|---|---|
| Operating voltage: | 5 – 20 VDC |
| Lens size sensor: | 23 mm (Default) |
| Block time: | 2.5 s |
| Angle sensor: | <110 cone angle |



### Purpose:

*Detects motion by sensing changes in infrared radiation*

# METHODOLOGY

## HARDWARE COMPONENTS

### Microcontroller

| Model Name/ No. | ESP32 |
|---|---|
| GPIO: | 36 Ports |
| ADC & DAC: | 14 & 2 Ports |
| Operating voltage: | 2.2 V – 3.6 V |
| Wi-Fi | 2.4 GHz |



### Purpose:

*Main controller (IR gateway) of the system*

# METHODOLOGY

## HARDWARE COMPONENTS

### Wireless transmitter module

| Model Name/ No. | ESP8266 Node MCU |
|---|---|
| GPIO: | 17 Ports |
| Peripheral Bus: | UART/SDIO/SPI/I2C/IR |
| Operating voltage: | 3.0 V – 3.6 V |
| Wi-Fi: | Station/ Soft AP/ Soft AP+Station |



### Purpose:

*Transmitter of gathered data signals from temperature, humidity, and motion sensors*

# METHODOLOGY

## HARDWARE COMPONENTS

### Push button switch

| Model Name/ No. | Tactile Push Button |
|---|---|
| Operation: | Tactile Feedback |
| Voltage: | 24 Vdc |
| Insulation Resistance: | 100 Mohms at 100 V |
| Operating Force: | 2.55 -+ 0.69 N |



### Purpose:

*Switch designed to temporary electrical circuit activation for user's voting input*

# METHODOLOGY

## HARDWARE COMPONENTS

### OLED Display

| Model Name/ No. | OLED I2C Display |
|---|---|
| Type: | Graphic |
| Display Format: | 128 x 64 dots |
| Controller: | SSD1306BZ |
| Duty Cyle: | 1/64 |




### Purpose:

*Display the command to be sent to the aircon*

# METHODOLOGY

## HARDWARE COMPONENTS

### Portable Power Bank

| Model Name/ No. | Rocoren Power bank RCBP45P-12 |
|---|---|
| Type: | Lithium Battery |
| Capacity: | 4500mAh |
| USB C Input | 5V |
| USB C Oputput | 5V |




### Purpose:

*Supply power to the sensor nodes and control hub*

# METHODOLOGY

## HARDWARE COMPONENTS

### Air-conditioning unit

| Model Name/ No. | Panasonic CW-XU1021VPH |
|---|---|
| Cooling Capacity: | 1.0 HP |
| Refrigerant: | R32 |
| Gross weight: | 40 kg |
| EER: | 12.4 kJ/hW |



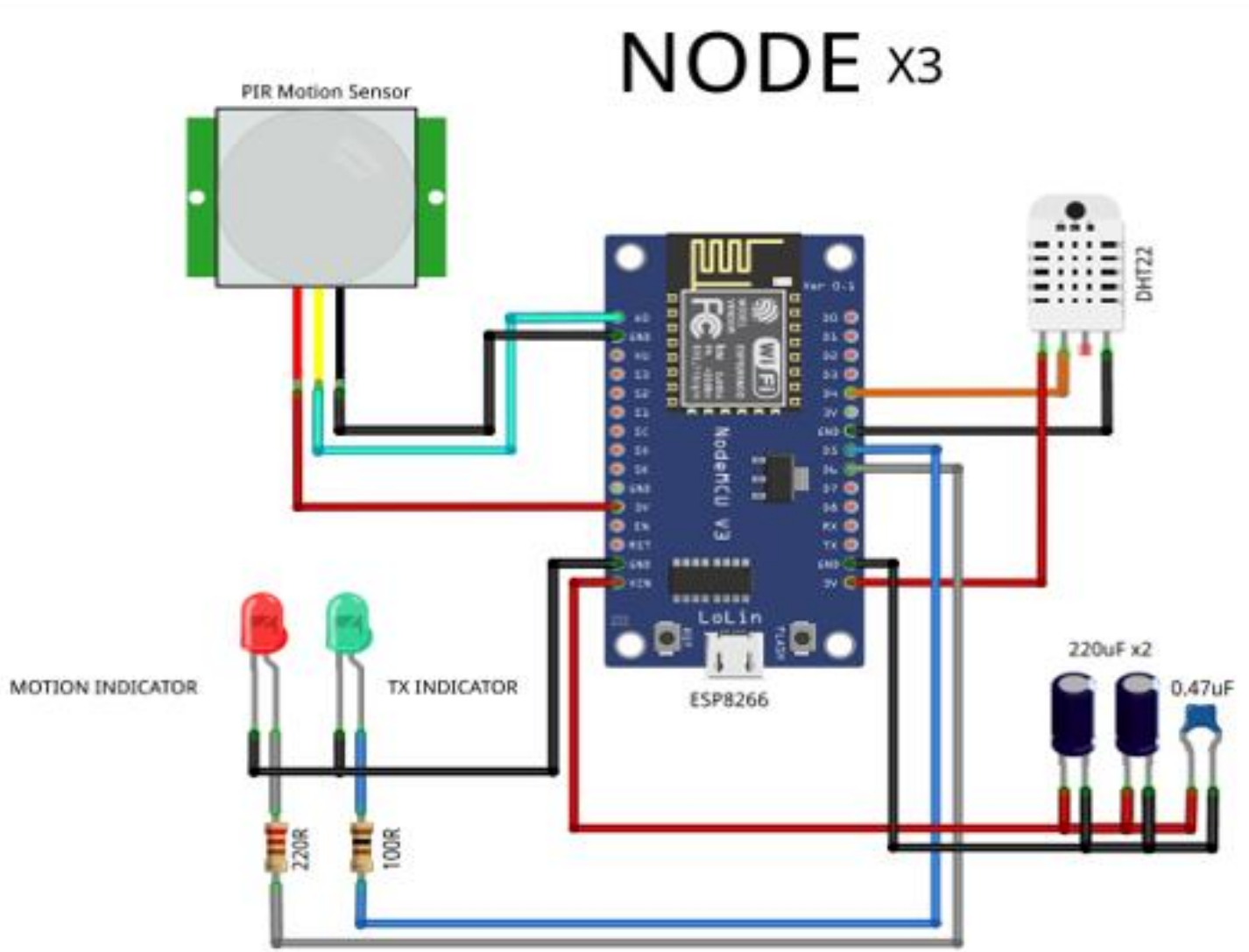

### Purpose:

*The receiver of the signal data transmitted by the MCU*

# METHODOLOGY

## SCHEMATIC DIAGRAM

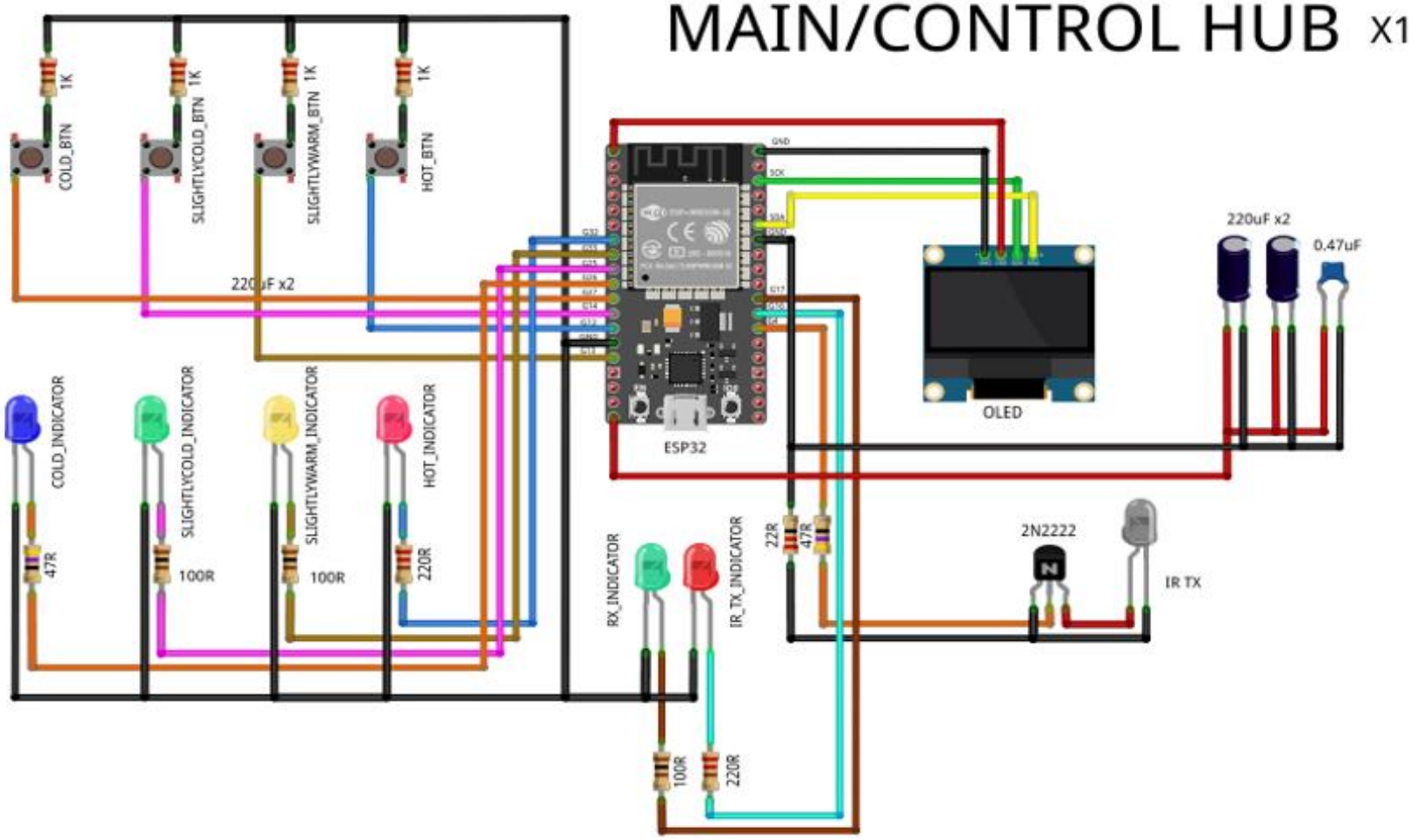
Sensor nodes



# METHODOLOGY

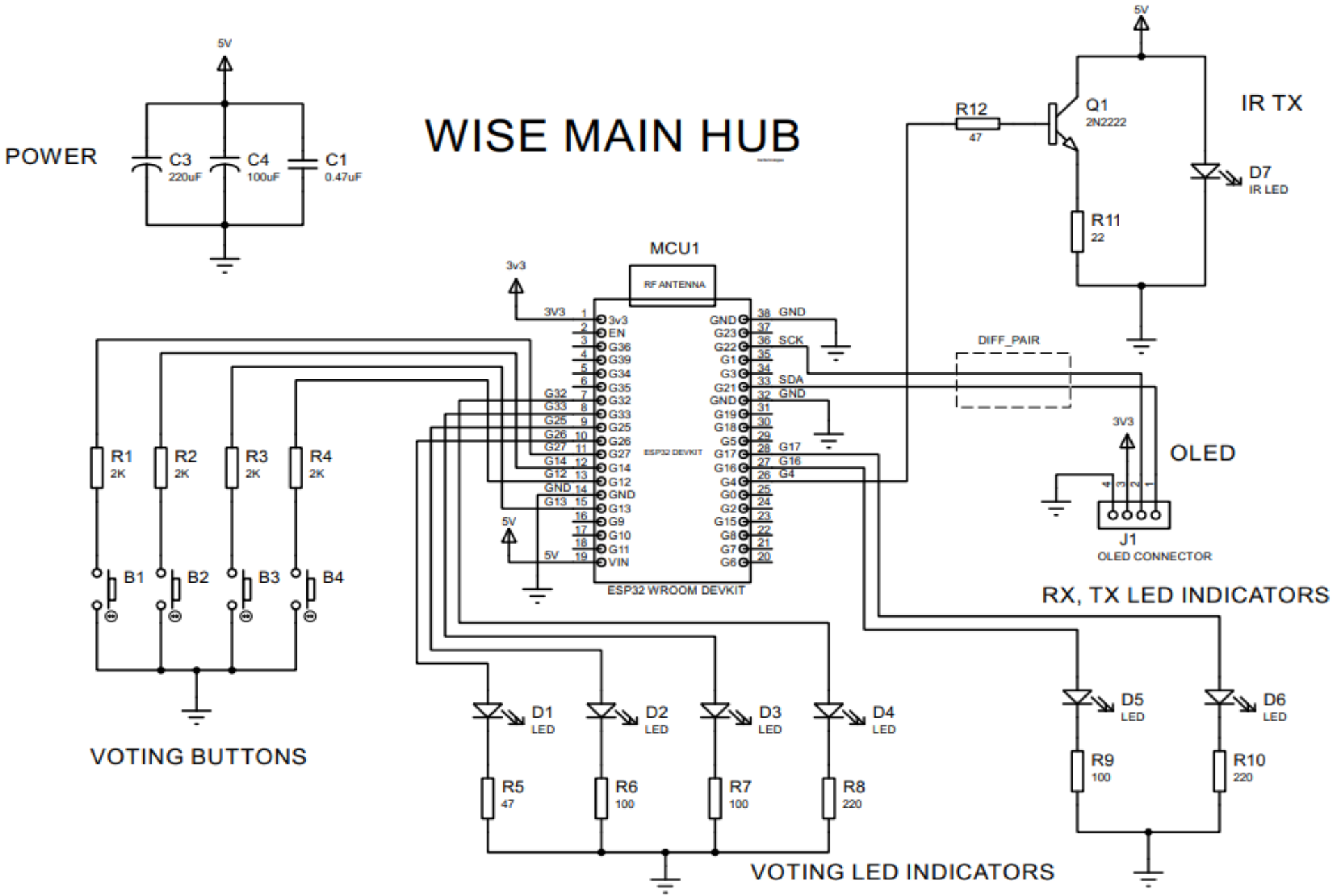
## SCHEMATIC DIAGRAM

Control Hub



# SCHEMATIC DIAGRAM

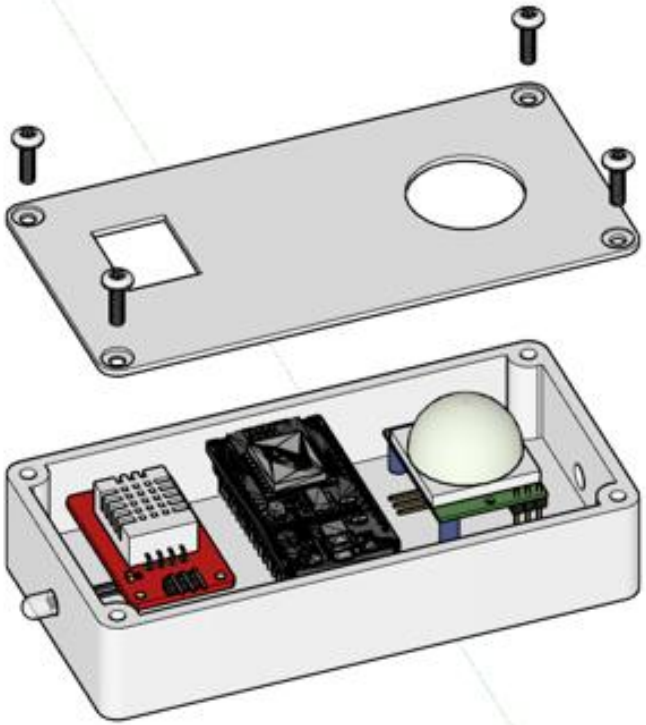
Control Hub

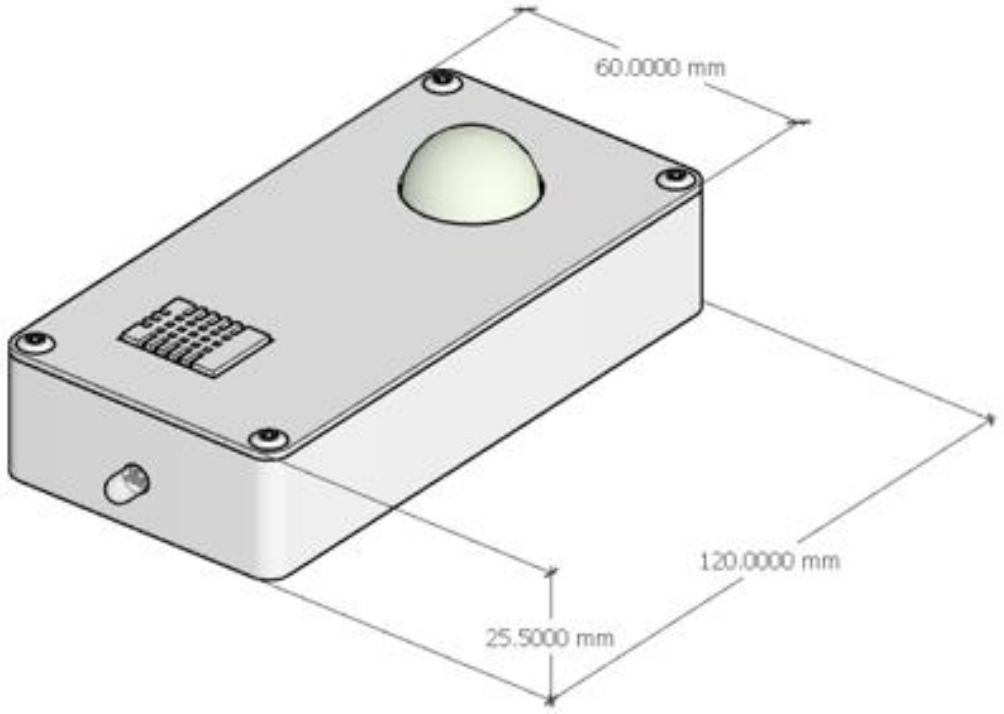


# METHODOLOGY

## SYSTEM MODEL

Sensor nodes





# METHODOLOGY

## SYSTEM MODEL

Control Hub



# EXPERIMENTS

**Exp 1 – Temperature and Humidity Accuracy Test**

**Exp 2 – Motion Sensor and Accuracy Test**

**Exp 3 – Wireless Sensor Nodes Performance and Data Accuracy Test**

# TEMPERATURE AND HUMIDITY ACCURACY TEST

**Objective:**
Measure the accuracy of DHT22 temperature and humidity sensor

**Materials:**

- 3 pcs - DHT22 temperature and humidity sensor
- 1 unit – Digital thermometer
- 1 unit – Hygrometer
- 1 unit – Panasonic Window Type air-conditioning unit
- 1 lot – 5m x 7m Airconditioned classroom
- 1 file – DHT22_Calibration.ino

**Procedures**

1. Prepare the workstation and the materials needed.
2. Prepare the software for the DHT22 accuracy test:
   a. Power on the laptop
   b. Once the laptop is powered on and logged in, locate the Arduino IDE software icon on the desktop and double-click it to open.
   c. In the Arduino IDE, go to File Tab, click it and navigate to the project folder, and select the file named "**DHT22_calibration.ino**".
   d. Connect the ESP8266 Node MCU with DHT22 sensor to the laptop using a USB data cable.
   e. In the Arduino IDE, go Tools then Port and verify that the laptop recognizes the connected COM port.

3. Upload the calibration firmware:

a. After the ESP8266 Node MCU successfully connected to the laptop.

b. Click the right arrow icon (Upload) in the Arduino IDE toolbar.

c. Check the status bar at the bottom of the window; and wait for the text "Done Uploading".

d. Upon successful upload, click the magnifying glass icon to open the serial monitor. Keep the device run for 3 minutes for system calibration.

e. While waiting, position the thermometer and hygrometer directly proportional to each other with 3 cm apart.

4. After 3 minutes of system calibration of the DHT22 sensor and microcontroller. Turn on the digital thermometer and hygrometer.

5. Once the DHT22 sensor, thermometer and hygrometer are all functioning properly, record all output readings. Enter the temperature readings from the DHT22 sensor and the thermometer in **Table 3.1** under the row labeled "**Temperature Readings**". Record the humidity readings from the DHT22 sensor and the hygrometer in **Table 3.2** under the row labeled "**Humidity Readings**" .

6. Compute the final average value of all the DHT22 sensor readings using:

$$\bar{x} = \frac{\sum_{i=1}^{n} X_i}{n} \qquad (3.1)$$

where:

$\bar{x}$ = average value of DHT22 readings

$x_i$ = temperature reading of DHT22 sensor

$n$ = number of trials

7. Calculate the **Mean Absolute Error (MAE)**, to determine the differences between the DHT22 sensor and thermometer using:

$$MAE = \frac{\sum_{i=1}^{n} |S_i - T_i|}{n} \qquad (3.2)$$

where:

***MAE*** = average physical deviation in °C

$S_i$ = sensor reading for trial i

$T_i$ = thermometer reading for trial i

8. Calculate the **Mean Percent Error (MPE**), to determine the average error relative to the actual temperature:

$$MPE = \frac{1}{2}\sum_{i=1}^{n}\left(\frac{|S_i - T_i|}{T_i}\right) \times 100 \qquad (3.3)$$

where:

***MPE*** = average percentage of error across all trials

$\frac{|S_i - T_i|}{T_i}$ = relative error for one trial

9. Compute for the **overall accuracy (A)** to determine the final system performance score using:

$$A = 100 - MPE \qquad (3.4)$$

where:

***A*** = the percentage of trials that match the reference standard (%)

10. Calculate the **Coefficient of Variation (CV)** to determine the precision and consistency of the DHT22 sensor using:

$$CV = (\frac{\sigma}{\bar{x}}) \qquad (3.5)$$

where:

**σ** = *Standard deviation*

Record summary metrics in **Table 3.3** in their respective columns and rows.

8. Repeat the process for 100 trials in the same workstation.

Table 3.1: Temperature Readings

| Trial #(n) | DHT22 Sensor Output (°C) | Digital Thermometer Output (°C) |
|---|---|---|
| 1 | 25.25°C | 24.82°C |
| 2 | 24.93°C | 24.80°C |
| 3 | 25.32°C | 25.22°C |
| ... | 25.76°C | 25.22°C |
| 100 | 24.88°C | 24.83°C |

Table 3.2: Humidity Readings

| Trial #(n) | DHT22 Sensor Output (%) | Hygrometer Output (%) |
|---|---|---|
| 1 | 56.07% | 54.42% |
| 2 | 56.68% | 55.56% |
| 3 | 58.25% | 58.25% |
| ... | 58.16% | 59.38% |
| 100 | 50.87% | 50.83% |

Table 3.3: Summary of Calibration Accuracy Metrics in Temperature

| Test | Results |
|---|---|
| Mean Absolute Error (MAE) | |
| Mean Percent Error (MPE) | |
| Overall Accuracy (A) | |
| Sensor Consistency | |

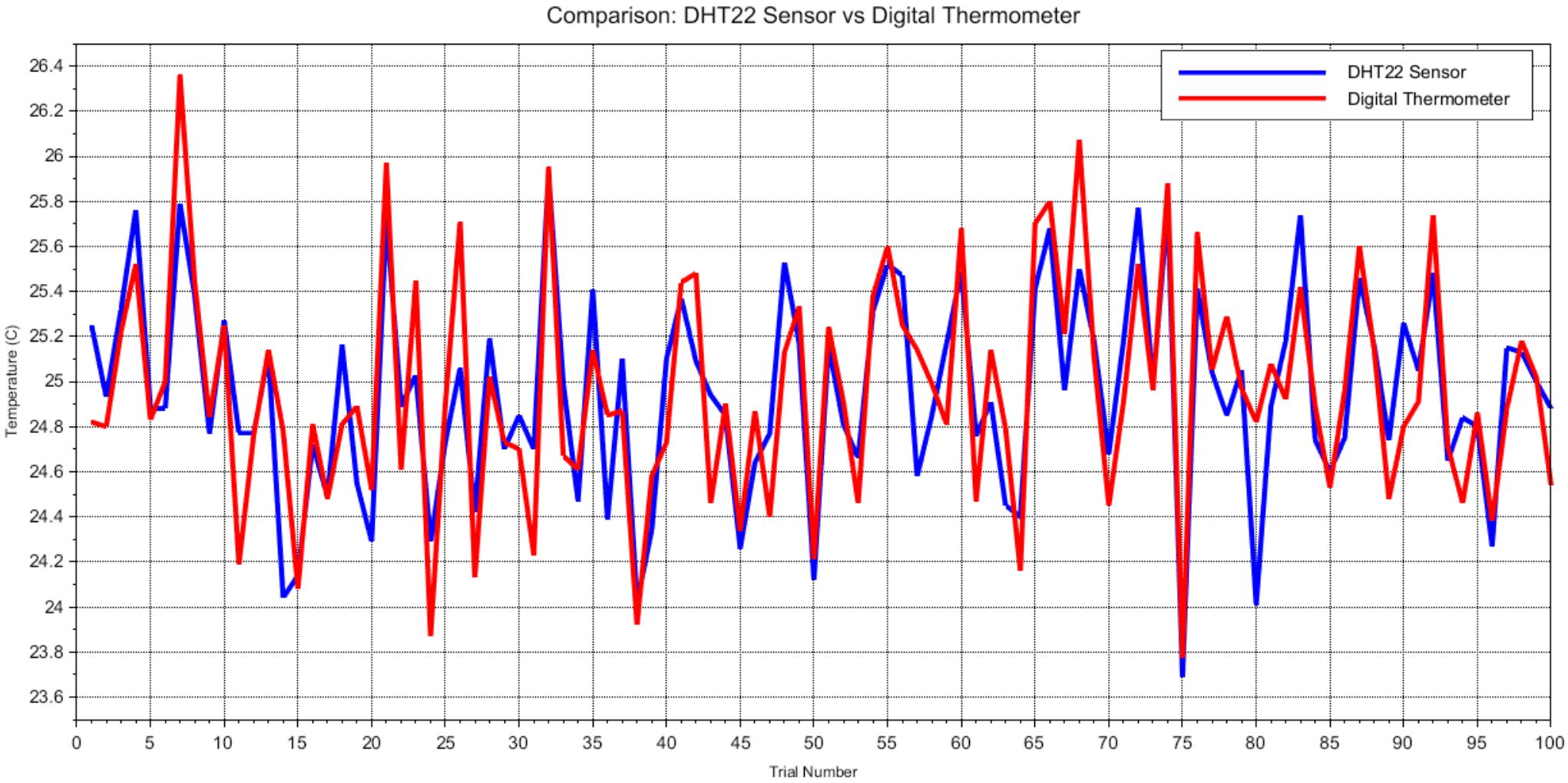
11. Repeat the process and calculations for humidity readings.

Table 3.4: Summary of Calibration Accuracy Metrics in Humidity

| Test | Results |
|---|---|
| Mean Absolute Error (MAE) | |
| Mean Percent Error (MPE) | |
| Overall Accuracy (A) | |
| Sensor Consistency | |

# 4.1 Temperature Readings

Figure 4.1: Comparison: DHT22 Sensor and Digital Thermometer



# 4.2 Humidity Readings

Figure 4.2: Comparison: DHT22 Sensor and Hygrometer

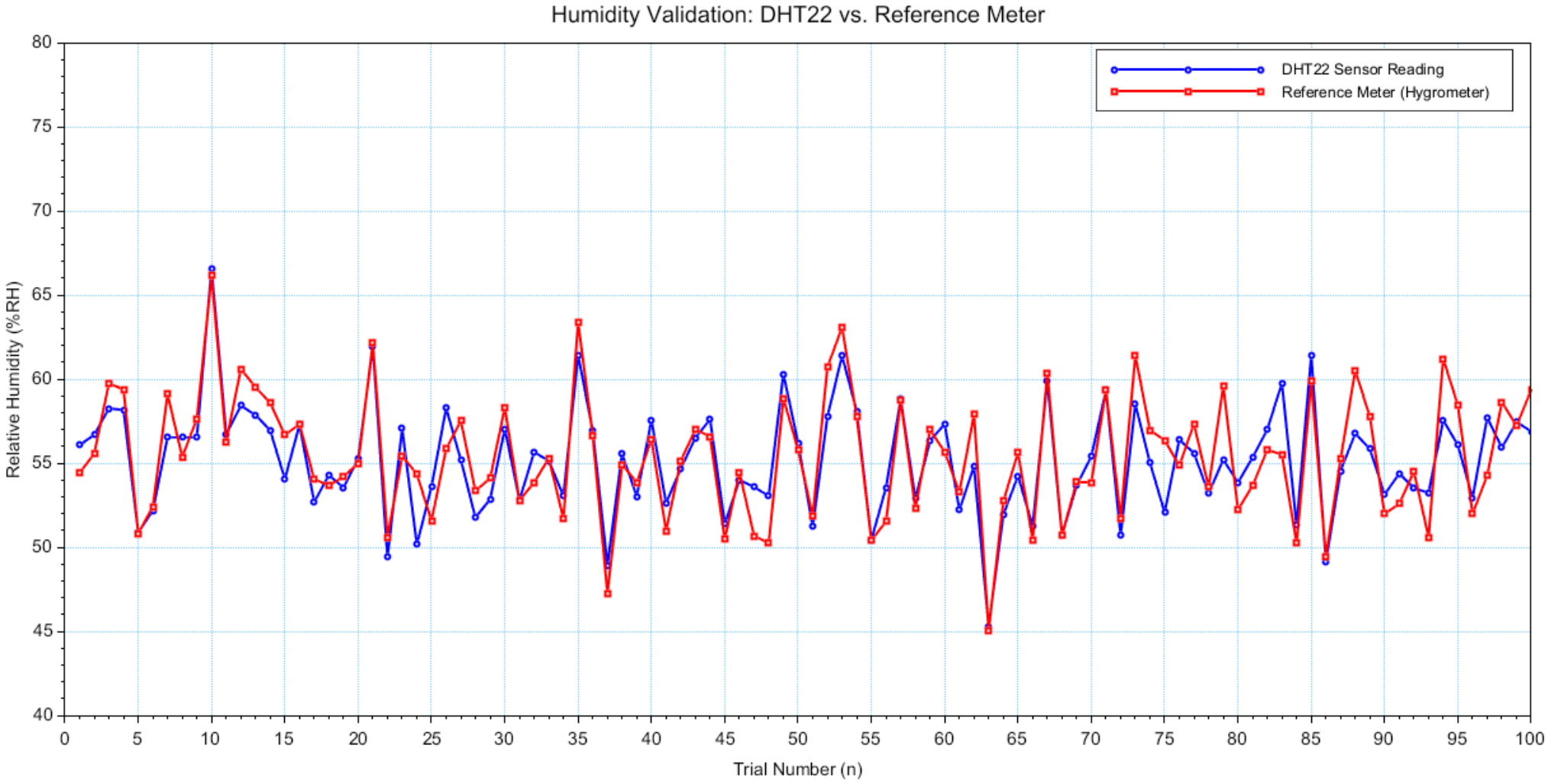


Table 3.3: Summary of Calibration Accuracy Metrics in Temperature

| Test | Results |
|---|---|
| Mean Absolute Error (MAE) | 0.23°C |
| Mean Percent Error (MPE) | 0.91% |
| Overall Accuracy (A) | 99.09% |
| Sensor Consistency (CV) | 98.18% |

Table 3.4: Summary of Calibration Accuracy Metrics in Humidity

| Test | Results |
|---|---|
| Mean Absolute Error (MAE) | 1.42% RH |
| Mean Percent Error (MPE) | 2.54% |
| Overall Accuracy (A) | 97.46% |
| Sensor Consistency | 94.11% |

**Key metric (CV):** The Coefficient of Variation (CV) is **1.82%** for the temperature readings; for the humidity readings the sensor exhibited a CV of **5.89%** while the hygrometer showed a CV of **6.68%**. These results indicate high stability in the data collection process.

# MOTION SENSOR ACCURACY TEST

## Objective

To measure the detection accuracy and reliability of HCSR501 (PIR) sensor

## Materials

- 3 pcs. HCSR501 PIR motion sensor
- 1 pc. ESP8266 Node MCU
- 1 pc. Protractor
- 1 lot. Air-conditioned room (Dimensions: 5m x 7m)
- 1 file. HCSR501_calibration.ino

**Procedures**

1. Prepare the workstation and the materials needed.
2. Prepare the software for the HCSR501 accuracy test:
   a. Power on the laptop
   b. Once the laptop is powered on and logged in, locate the Arduino IDE software icon on the desktop and double-click it to open.
   c. In the Arduino IDE, go to File Tab, click it and navigate to the project folder, and select the file named "HCSR501_calibration.ino".
   d. Connect the ESP8266 Node MCU with HCSR501 sensor to the laptop using a USB data cable.
   e. In the Arduino IDE, go Tools then Port and verify that the laptop recognizes the connected COM port.

3. Upload the calibration firmware:

a. After the ESP8266 Node MCU successfully connected to the laptop.

b. Click the right arrow icon (Upload) in the Arduino IDE toolbar.

c. Check the status bar at the bottom of the window; and wait for the text "Done Uploading".

d. Upon successful upload, click the magnifying glass icon to open the serial monitor. Keep the device run for 3 minutes for system calibration.

e. While waiting, Mount the HCSR501 sensor on the classroom wall/post at a height of 2.2m, facing the center of the 7m x 5m room.

4. Using a measuring tape or ruler, mark distances from the sensor at 1 m intervals, from 1 meter to 4 meters.

5. Establish angular reference lines relative to the sensor centerline; 0° directly in front of the sensor, ±15°,±30°, and ±45°.

6. Mark these angles on the floor using tape or protractor.

7. Establish the voltage-based decision rule for the motion detection.

8. If the PIR output voltage is **≥ 2.5 V**, label the output as HIGH (motion detected).

9. If the PIR output voltage **≈ 0 V**, label the output as LOW (no motion detected).

10. After labeling the distances and angles in the floor, position the test subject (person) at 1 m distance and 0° angle.

11. The test subject should walk perpendicular to the sensor's field of view at normal walking speed and expect a high output voltage.

12.Apply the decision rule consistently and record the labels (**HIGH or LOW**) in the **Table 3.4**

Table 3.4: Motion output at 1m;0°

| Trial # (n) | Motion High (V) | Motion Low (V) |
|---|---|---|
| 1 | | |
| 2 | | |
| ... | | |
| 100 | | |

13. Repeat the test 100 times at the same distance and angle to test accuracy and consistency.

14. Calculate the **success rate (S)** for every motion detected using;

$$S = \frac{M_d}{n} \times 100 \qquad (3.6)$$

where:

$S$ = success rate
$Md$ = high output motion (Motion detected)
$n$ = number of trials

15. Repeat steps 11-14 for all marked angles (±15°,±30°,±45°) at the same distance.

16. Record all the data from different angles at the same distance in **Table 3.5** to determine the highest detection rate of the sensor.

Table 3.5: Summary of detection rate at 1 meter distance

| Distance (m) | Angle (°) | Total Trials | Total Motion High | Detection Rate |
|---|---|---|---|---|
| 1 meter | 0 | 100 | | |
| | 15 | 100 | | |
| | 30 | 100 | | |
| | 45 | 100 | | |

where:

$S$ = success rate

$Md$ = high output motion (Motion detected)

$n$ = number of trials

17. Increase the distance to 2m, then repeat the process.
18. Continue testing for distances up 4 meters and record the data.

Table 3.5: Summary of detection rate at 1 meter distance

| Distance (m) | Angle (°) | Total Trials | Total Motion High | Detection Rate |
|---|---|---|---|---|
| 1 meter | 0 | 100 | 92 | 92% |
| | 15 | 100 | 95 | 95% |
| | 30 | 100 | 97 | 97% |
| | 45 | 100 | 90 | 90% |

Table 3.5.1: Summary of detection rate at 2 meters distance

| Distance (m) | Angle (°) | Total Trials | Total Motion High | Detection Rate |
|---|---|---|---|---|
| 2 meters | 0 | 100 | 96 | 96% |
| | 15 | 100 | 97 | 97% |
| | 30 | 100 | 98 | 98% |
| | 45 | 100 | 97 | 97% |

Table 3.5.2: Summary of detection rate at 3 meters distance

| Distance (m) | Angle (°) | Total Trials | Total Motion High | Detection Rate |
|---|---|---|---|---|
| 3 meters | 0 | 100 | 92 | 92% |
| | 15 | 100 | 95 | 95% |
| | 30 | 100 | 97 | 97% |
| | 45 | 100 | 90 | 90% |

Table 3.5.3: Summary of detection rate at 4 meters distance

| Distance (m) | Angle (°) | Total Trials | Total Motion High | Detection Rate |
|---|---|---|---|---|
| 4 meters | 0 | 100 | 92 | 92% |
| | 15 | 100 | 95 | 95% |
| | 30 | 100 | 97 | 97% |
| | 45 | 100 | 90 | 90% |

# WIRELESS SENSOR NODES PERFORMANCE AND DATA ACCURACY TEST

## Objective

To measure the accuracy, stability, and reliability of the three sensor nodes

## Materials

- 3 pcs. Sensor nodes
- 1 pc. Control hub
- 1 lot. Air-conditioned room (Dimensions: 5m x 7m)
- 1 file. SensorNodes_accuracy_test.ino

**Procedures:**

1. Prepare all materials needed for the experiment. Mount the three sensor nodes on the classroom wall at a height of **2.2 m**, place each node a fixed position (**Left, Center, Right post**)of the room.
2. For identification purposes assign sensor node ID; sensor node 1 (left part), sensor node 2(center), sensor node 3 (right part).
3. Open the Arduino IDE on the laptop and connect the ESP32 (Control Hub). Upload the receiver program and open the serial monitor to verify readiness for incoming data.
4. Power on sensor node 1 and allow the system to stabilize for 60 seconds with no movement inside the classroom

5. Define one transmission trial as the sensor node sending one complete data packet containing: **Node ID**, **Voltage output** (V) for motion, **Temperature** (T) in ℃ and **Humidity** (H) in **%**.

6. Start the transmission process for sensor node 1

7. Allow the node to transmit exactly 100 consecutive packets every 10 seconds.

8. For each packet if the control hub receives the complete packet, mark the trial as **successful**.

9. For each packet is missing or corrupted, mark the trial as **failed**.

10. Record the data packet consists of voltage output, temperature, and humidity in the **Table 3.6**

Table 3.6: Data Packets for Sensor Node 1

| Trial # (n) | Data Packet | | | Data Status (Successful/Failed) |
|---|---|---|---|---|
| | PIR Voltage (V) | Temperature (℃) | Humidity (%) | |
| 1 | | | | |
| … | | | | |
| 100 | | | | |

11. Calculate the **transmission success rate (TSR)** for each sensor node using;

$$TSR = \frac{N_{scs}}{100} \cdot 100 \qquad (3.7)$$

where:

$TSR$ = *transmission success rate (%)*

$N_{scs}$ = number of packets successfully received by the control hub

12. Record the transmission success rate for each sensor node in **Table 3.7**
13. Repeat the process for Sensor node 2 and 3

Table 3.7: Data Packets Summary of Sensor Nodes

| Sensor Node ID | Packets Sent | Packets Successfully Received | Failed Transmissions | Transmission Success Rate (%) |
|---|---|---|---|---|
| Sensor Node 1 | 100 | 87 | 13 | 87% |
| Sensor Node 2 | 100 | 80% | 20 | 80% |
| Sensor Node 3 | 100 | 92 | 8 | 92% |

# AIRWISE TRANSMISSION TO PANASONIC AIRCON; IR TEST

## Objectives

To measure the success rate that the AC performs the expected action when the system sends command;

To measure success rate over many repeated trials

## Materials

- 1 pc. AIRWise control system
- 3 pcs. Sensor Nodes
- 1 lot. Air-conditioned room (Dimensions: 5m x 7m)
- 1 unit. Pasasonic Window-type Aircon
- 1 file. Airwise_test.ino

## Procedures

1. Secure formal authorization from the appropriate school authorities to conduct the experimental study during regular class hours in **GLEC Room 1**. Briefing the assigned instructor to explain the scope, objectives, and duration of the experiment, including the 30-minute pre-class set-up period. Make sure to provide the written permission or access endorsement to confirm authorized use of the classroom and air-conditioning unit during the experiment.
2. Schedule the experimental set-up begin **30 minutes prior** to the scheduled class session. Ensure the GLEC Room 1 remains unoccupied during this period to prevent human activity from influencing baseline thermal conditions.

3. Prepare the classroom by closing all doors and windows (if applicable) and ventilation or external airflow is introduced. Keep the classroom equipment (if applicable) such as computers, projectors, TV in a constant state to minimize uncontrolled heat sources.

4. Prepare all the sensor nodes and the control hub.

5. Verify that the air-conditioning unit in GLEC room 1 is initially powered off. Allow the room's ambient temperature and humidity to stabilize before activating any control mechanisms.

6. Mount the three sensor nodes on the classroom wall at a height of **2.2 m**, place each node a fixed position (**Left, Center, Right post**)of the room.

7. For identification purposes assign sensor node ID; sensor node 1(left part), sensor node 2(center), sensor node 3 (right part). Ensure that each transmitted packet includes; Sensor Node ID, temperature, humidity, and motion readings.

8. Position the control hub on a stable surface exactly 2 meters from the air-conditioning unit. Align the control hub's IR transmitter directly toward the AC's IR receiver to maintain a clear Line-of-sight.

9. Prepare the software for the AIRWise test:

   a. Power on the laptop

   b. Once the laptop is powered on and logged in, find the Arduino IDE software icon on the desktop and double-click it to open.

a. In the Arduino IDE, go to File Tab, click it and navigate to the project folder, and select the file named "AIRwise_test.ino".

b. Connect the control hub to the laptop using a USB data cable.

c. In the Arduino IDE, go Tools then Port and verify that the laptop recognizes the connected COM port.

10. Upload the calibration firmware:

a. After the control hub successfully connected to the laptop.

b. Click the right arrow icon (Upload) in the Arduino IDE toolbar.

c. Check the status bar at the bottom of the window; and wait for the text"Done Uploading".

d. Upon successful upload, click the magnifying glass icon to open the serial monitor. Keep the device run for 3 minutes for system calibration.

11. Power on the control hub and verify successful reception and decoding of IR data packets transmitted by each sensor node. Monitor the data reception through serial output or diagnostic indicators.

12. Record baseline temperature and humidity readings from all sensor nodes for a fixed observation period of ten (10) minutes. During this baseline period, prevent the control hub from issuing any air conditioning control commands.

13. Allow the system to operate autonomously for the remaining pre-class period to verify stable communication between sensor nodes, the control hub, and the air conditioning unit.

14. Once the set-up and initializations were done. Permit students to enter Room GLEC 1 at the scheduled class time without modifying the system configuration. Continue system operation during class hours to evaluate performance under actual occupancy conditions.

15. Continuously collect temperature and humidity data from each sensor node during class occupancy. Monitor how environmental conditions change in response to student presence and activity.

16. Assess infrared communication reliability by recording the number of data packets successfully received by the control hub relative to the total number of packets transmitted by each sensor node.

17. Record and evaluate the effectiveness of infrared communication between the control hub and the air conditioning unit by determining whether the data packets transmitted by the control hub is proportional to the data packets successful received and executed by the air conditioner in the **Table 3.8** and **Table 3.9**

Table 3.8: Data Packets Sent from Control Hub to Aircon

| Trial # (n) | Mode | Temperature | Swing (On/Off) | AC (On/Off) |
|---|---|---|---|---|
| 1 | | | | |
| 2 | | | | |
| ... | | | | |
| 100 | | | | |

Table 3.9: Data Packets received from Control hub

| Trial # (n) | Mode | Temperature | Swing (On/Off) | AC (On/Off) |
|---|---|---|---|---|
| 1 | | | | |
| 2 | | | | |
| ... | | | | |
| 100 | | | | |

17. Compute the proportionality ratio to assess transmission reliability and command execution consistency using;

$$P_{IR} = \frac{N_{exec}}{N_{sent}} \qquad (3.8)$$

where:

$P_{IR}$ = IR Communication proportionality ratio

$N_{exec}$ = Number of IR commands successfully received and executed

$N_{sent}$ = Number of IR commands transmitted by the control hub

# EXPERIMENT 5
## VOTING MECHANISM ACCURACY TEST

### Objective

To evaluate whether the voting mechanism can correctly adjust the AC set temperature based on different vote combinations by comparing the expected and actual $T_{snew}$ for each voting behavior test cases

### Materials

- 1 pc.  ESP32
- 1 pc. IR Tx
- 3 pcs. Sensor nodes
- 1 lot. Air-conditioned room (Dimensions: 5m x 7m)
- 1 unit. Pasasonic Window-type Aircon
- 1 file. VM_accuracy_test.ino

# EXPERIMENT 5
## VOTING MECHANISM ACCURACY TEST

**Procedures:**

1. Prepare all materials: ESP32, IR transmitter, voting buttons (Warm, Slightly Warm, Slightly Cool, Cool), laptop with Arduino IDE, and 3 sensor nodes.
2. Set the air conditioner to Cool mode and stabilize at initial baseline temperature $T_b$ of 22°C.
3. Connect the ESP32 to the laptop, open Arduino IDE, load the voting mechanism program, and open the serial monitor to view output.
4. For each test case in Table 3.6, set $T_b$ to 22°C or 24°C as required and allow the air conditioner to stabilize.
5. Start the 5-minute voting session by pressing the first button, press buttons according to the selected vote pattern (W, SW, SC, C) from Table 3.6.

# EXPERIMENT 5
## VOTING MECHANISM ACCURACY TEST

**Procedures:**

6. After completing the pattern, do not press any more buttons, wait until the full 5minutes has elapsed and the program finishes the voting session.

7. When the session ends, record Actual $T_{snew}$ from serial monitor in Table 3.6 and confirm that $5 \leq N \leq 20$, discard invalid trials (N outside 5-20) before recording data.

8. Using $T_b$ and $\Delta T$ from the serial monitor, calculate Expected $T_{snew}$ with equation (3.4)and record in Table 3.6

9. Repeat steps 4 to 8 for 100 trials per test case.

10. For each case, decide Result: Pass if ≥95 out of 100 trials show correct direction(Warm-dominant: $T_{snew} > Tb$; Cool-dominant: $T_{snew} < Tb$; Mixed: $T_{snew} \approx Tb$)and Actual $T_{snew}$ is within $\pm 0.5°C$ of Expected $T_{snew}$; otherwise Fail. Record in Table 3.6.

# EXPERIMENT 5
## VOTING MECHANISM ACCURACY TEST

$$T_{\text{snew}} = T_b + \Delta T \tag{3.4}$$

where:

$T_b$ = Baseline temperature (°C)

$\Delta T$ = Temperature change (°C)

# EXPERIMENT 5
## VOTING MECHANISM ACCURACY TEST

| Case | Total Trials | Tb (°C) | Vote Combination (W, SW, SC, C) | Expected $T_{snew}$ (°C) | Actual $T_{snew}$ (°C) | Result (Pass/Fail) |
|---|---|---|---|---|---|---|
| 1 | 100 | 22 | 5, 0, 0, 0 | | | |
| 2 | 100 | 22 | 0, 0, 0, 5 | | | |
| 3 | 100 | 22 | 3, 3, 2, 2 | | | |
| 4 | 100 | 24 | 3, 3, 2, 2 | | | |
| 5 | 100 | 24 | 20, 0, 0, 0 | | | |